YupReserved

A hotel reservation system for India

Database Management Systems — project log, report and viva guide

PostgreSQL 16 · 15 tables · 408,202 rows · FastAPI · React

Submitted by: _____ Enrolment no.: _____

Course: _____ Guide: _____

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Part A · Work log

This part is updated after every working session, so the whole history of the project can be traced: what was built, which files were added, which tables and concepts were involved, and why each decision was taken.

Session 1 — 29 October 2026: Project foundation: dataset, database, backend, CLI, frontend and this report

Built the complete YupReserved system from an empty folder. A real public dataset of 4,000 Indian hotel listings was downloaded, cleaned and normalised into a 15-table PostgreSQL database with 408,192 rows; a FastAPI backend, a terminal CLI and a React frontend were built on top of it, and every syllabus unit was mapped to a working, demonstrable feature.

Decisions and why

PostgreSQL 16 instead of the installed MySQL 5.7. MySQL 5.7 parses CHECK constraints but does not enforce them, and has no FULL OUTER JOIN, INTERSECT/EXCEPT, CTEs or window functions — all needed for the syllabus. PostgreSQL also offers EXCLUDE constraints, deferred constraint triggers, table inheritance (OODB), partitioning (distributed fragments), full-text search (digital libraries) and readable EXPLAIN ANALYZE plans.

Goibibo hotel listings (Kaggle, PromptCloud) as the base dataset. 4,000 real Indian properties with city, state, star rating, coordinates, facilities, nearby points of interest and room counts. It contains genuine 1NF violations (pipe-separated lists) and inconsistent state spellings, which make normalisation demonstrable on real data.

Prices generated by a documented pricing model. No public dataset of Indian hotels includes nightly prices (the Cities42 price file is not publicly downloadable). Base prices are computed from star rating, property type and city, and every booking is priced by the same SQL function the website uses. This is stated openly in the dataset README.

Exactly 15 core tables, teaching objects kept in separate schemas. The 15 business tables live in schema public. Raw data (schema staging) and demonstrations (schema lab: normal-form walk-through, OODB, fragments, multimedia, transaction demo table) never mix with business data.

Surrogate room_id, and rooms left in 3NF (not BCNF) on purpose. With room_type_id → hotel_id, BCNF decomposition would lose the dependency (hotel_id, room_number) → room_id. 3NF keeps every dependency enforceable; a composite foreign key keeps room and room type in the same hotel.

Raw SQL, no ORM; every API response carries its SQL. So that every screen can be explained in the viva: the frontend's 'Under the hood' drawer shows the exact SQL, rows and timing behind each page.

Server and Python/Node tooling stored outside the project. PostgreSQL binaries, data directory and the Python virtual environment live in ~/.yupreserved so the project folder stays clean and presentable in the file explorer.

Files added or changed

file / folder	what it is
01-dataset/raw/goibibo_com-travel_sample.csv	the untouched source dataset (4,000 listings, 36 columns)

file / folder	what it is
01-dataset/reference/*.csv, signature_stays.json	states (ISO-style codes), festival seasons, 95 curated landmarks with coordinates, 18 original YupReserved stays
01-dataset/pipeline/vocab.py	cleaning vocabularies: state aliases, city fixes, chain-brand filter, 356 → 52 amenity mapping
01-dataset/pipeline/build_dataset.py	clean → enrich → simulate pipeline producing one CSV per table
01-dataset/processed/*.csv	14 clean CSV files, 329,508 rows (+ manifest and amenity mapping)
02-database/build.sql	rebuilds the whole database in one command
02-database/schema/01…07_*.sql	domains & enums, 15 tables, functions/procedures, 8 triggers, indexes, views, roles
02-database/seed/load_data.sql	bulk load with \copy; prices and payments derived in SQL
02-database/lab/lab_objects.sql	raw staging table, UNF→3NF tables, 4NF/5NF demos, OODB, fragments, media, txn demo
02-database/queries/01…07_*.sql	66 named, annotated queries (relational algebra + explanation)
02-database/demos/why_not.sql, query_processing.sql	16 constraint scenarios and 15 query-plan demonstrations
03-backend/app/*	FastAPI app: db layer, catalog parser, error explainer, lab engine, concurrency demos, 3 routers
04-cli/yup.py and ./yup	terminal tool: tables, show, describe, keys, q, explain, why, txn, quote, find, sql, recovery
05-frontend/src/*	React + TypeScript site: poster-art engine, 8 pages, design system, EN/□□□, light/dark
06-docs/diagrams/*	ER (Chen), relational schema, architecture and workflow diagrams (Graphviz)
06-docs/report/*	this report: work log + builder
scripts/db.sh	start / stop / build / psql helper for the local PostgreSQL server

Tables involved

states, cities, landmarks, hotels, hotel_landmarks, amenities, hotel_amenities, room_types, rooms, seasons, guests, bookings, booking_rooms, payments, reviews

DBMS concepts used

- ER model: strong & weak entities, partial key, 1:1 / 1:N / M:N, recursive relationship, multivalued & derived attributes
- Keys: super, candidate, primary, alternate, composite, surrogate, natural, foreign (incl. self-referencing and composite)
- Integrity: domains, NOT NULL, CHECK, UNIQUE, EXCLUDE, referential actions (RESTRICT / CASCADE / SET NULL)
- Triggers (BEFORE / AFTER / deferred constraint trigger), stored functions and a stored procedure
- Views, updatable view WITH CHECK OPTION, security view, materialized view
- Relational algebra (σ π ρ ⋈ ∪ ∩ − ÷ γ), tuple and domain relational calculus
- All joins: inner, natural, theta, left/right/full outer, self, cross, semi, anti, lateral, multi-way
- Functional dependencies, 1NF → 2NF → 3NF → BCNF, MVD / 4NF, join dependency / 5NF, lossless join
- Query processing: seq/index/bitmap scans, hash/merge/nested-loop joins, external sort, pipelining, heuristics

- Transactions: ACID, 2PL row locks, deadlock detection, isolation levels, MVCC, WAL crash recovery
- Distributed (fragmentation), object-relational (types, inheritance), multimedia (bytea), data mining, full-text search
- DCL: roles, GRANT / REVOKE, least privilege, SECURITY DEFINER

Verified

- Database rebuilt from scratch with every trigger and constraint active: all 408,192 rows accepted.
- All 66 named queries run without error (all under 100 ms).
- All 16 'why not' scenarios are refused by the intended rule.
- All 15 query-plan demonstrations show the intended operator (e.g. external merge sort, merge join).
- All 5 concurrency demonstrations behave as expected (lock wait, deadlock victim, non-repeatable read, MVCC, atomicity).
- End-to-end booking, double-cancel refusal and capacity rejection tested through the website's API.
- Crash recovery demonstrated (./yup recovery): committed row survived, uncommitted row gone; API reconnects automatically.
- Lossless-join check on the cleaned data: 57,088 rows in, 57,088 out, 0 spurious (raw data was lossy: 357 rows).
- Bugs found and fixed during testing: double cancellation was allowed (now refused by sp_cancel_booking); window.scrollTo returning a Promise broke React effects; lab scenarios hit privileges instead of constraints (yup_lab role added).
- Frontend type-checks and builds for production (~140 kB gzipped JavaScript).

Session 2 — 29 October 2026: Fix: navigation, Back button, mood cards and 'Try' chips stopped responding

The user reported that the Back button, the header links (Stays, My trips, Under the hood), the mood cards and the 'Try' chips did nothing. Reproduced in a real Chrome browser with an automated test, found the single root cause, fixed it in three layers, and verified every link and a full booking end to end.

Decisions and why

Root cause: one page error crashed the whole app. Opening /stays without dates (every one of those links) passed 'undefined' dates into the search bar, which replaced the default dates; date arithmetic then threw 'Invalid time value'. With no error boundary, React unmounted the entire app, so no link or the Back button could respond any more.

Fix 1 — the cause. The search bar now ignores unset initial values, and the date helpers (fromISO/addDays) fall back to the platform's 'today' instead of producing an Invalid Date.

Fix 2 — a safety net. An ErrorBoundary around the pages, keyed by the URL path: if a page ever fails, a friendly message appears, while the header, Back and every other page keep working.

Fix 3 — feedback that survives a re-render. After cancelling, a trip moves from 'Coming up' to 'Past & cancelled' and its card is re-created, which erased its success message. The message is now a page-level banner.

Files added or changed

file / folder	what it is
05-frontend/src/components/search.tsx	ignore undefined initial values
05-frontend/src/lib/format.ts	safe date parsing
05-frontend/src/components/ErrorBoundary.tsx	new — page-level error boundary
05-frontend/src/App.tsx	routes wrapped in the error boundary
05-frontend/src/pages/Trips.tsx	page-level notice for cancellations and reviews
05-frontend/src/components/feedback.tsx	empty state can be the page's main heading (h1)
05-frontend/src/art/Poster.tsx	missing React list key

Tables involved

bookings, payments (two test bookings created and cancelled through the website)

DBMS concepts used

- Frontend reliability: error boundaries, defensive parsing of URL parameters
- The cancellation path exercised the stored procedure sp_cancel_booking and the refund trigger

Verified

- Automated browser test: all 15 entry points (3 header links, 8 mood cards, 4 'Try' chips) open their page and Back returns home — 15/15.
- Stays → hotel → Back → Back works.
- Full booking in the browser: hotel → Reserve → OTP sign-in → UPI → confirmation → My trips → cancel with refund banner.
- Type-check and production build pass.

Session 3 — 29 October 2026: Photos, HD hero, secure OTP, easier filters, installable mobile app, full-action CLI, deployment

Added real photos for 97% of hotels (Wikimedia Commons, free licences, credited), a rotating HD hero of 12 Indian landmarks, a genuine OTP sign-in with hashed codes, expiry, attempt and rate limits and signed session tokens, redesigned filters, a Progressive Web App with a phone tab bar, a CLI that can perform every action of the website, and one-container deployment files.

Decisions and why

Photos of the place, not of the hotel. The dataset has no hotel photos, and other sites' photos are copyrighted. Each hotel shows its nearest photographed landmark within 4 km, otherwise one of its city's photos (chosen by hotel_id for variety), always captioned and credited. Photos are hot-linked from Wikimedia, so nothing heavy is stored.

Photo columns, not a photo table. photo_url, photo_url_hd and photo_credit are attributes of cities and landmarks (each FD city_id → photo holds), so the core stays at exactly 15 tables. The view v_hotel_photo derives a hotel's photo.

A NULL-logic bug caught by the database. The first CHECK (photo_url ~ '^https://' AND photo_credit IS NOT NULL) rejected cities WITHOUT a photo: NULL AND FALSE is FALSE. Fixed to (photo_url IS NULL OR (…)). A real example of SQL three-valued logic.

OTP done properly. Codes come from a CSPRNG, only an HMAC-SHA256 is stored (schema auth, outside the 15 tables), 5-minute expiry, 5 attempts, 30 s resend cooldown, 3 codes per 10 minutes, SELECT … FOR UPDATE on the challenge. Sign-in returns an HMAC-signed token; bookings, trips, cancellations and reviews now check the token's guest, closing a hole where any guest_id could be sent. SMS providers (Fast2SMS, Twilio) plug in through environment variables; without one, demo mode shows the code on screen.

Progressive Web App instead of a separate native app. One codebase installs on Android and iPhone home screens (manifest, icons, service worker with offline cache), with a native-style bottom tab bar and safe-area support.

One container for deployment. FastAPI serves the built React site from the same origin (no CORS), packaged in a Dockerfile with a Render blueprint; deploy-db.sh copies the database to any hosted PostgreSQL with fresh random role passwords.

Files added or changed

file / folder	what it is
01-dataset/pipeline/fetch_photos.py, reference/photos.csv	Wikipedia/Commons API fetcher (batched, polite) and its cached result
02-database/schema/02_tables.sql, 06_views.sql	photo columns + CHECKs, view v_hotel_photo
02-database/schema/08_auth.sql	schema auth: otp_challenges
03-backend/app/security.py	OTP generation/hashing, SMS providers, signed session tokens
03-backend/app/routers/booking_routes.py	real OTP endpoints, ownership checks on every guest action
03-backend/app/main.py	serves the built website in production
04-cli/yup.py, 04-cli/README.md	search, hotel, avail, signup, book, booking, cancel, review, guest, trips, stats, menu, install, sql --write
05-frontend/src/components/Photo.tsx, AuthDialog.tsx	photo with poster fallback, HD hero slideshow; new sign-in dialog
05-frontend/src/pages/Search.tsx	redesigned filters
05-frontend/public/manifest.webmanifest, sw.js, icons/	installable app
Dockerfile, .dockerignore, render.yaml, scripts/deploy-db.sh	deployment

Tables involved

cities (+photo columns), landmarks (+photo columns), guests, bookings, payments, reviews, auth.otp_challenges

DBMS concepts used

- Three-valued logic in CHECK constraints (NULL AND FALSE = FALSE)
- Derived data through a view with LATERAL joins (v_hotel_photo)
- Security: hashing secrets, constant-time comparison, rate limiting, authorization by ownership
- Row locking (FOR UPDATE) on the OTP challenge so two verifications cannot race

Verified

- Database rebuilt from scratch with the new columns and schema: all rows accepted.

- OTP: wrong code counts attempts, codes cannot be reused, resend cooldown enforced, tampered token rejected, a guest cannot read or cancel another guest's bookings.
- Browser test: full booking with the new OTP (sign in → UPI → confirmation → cancel) and 15/15 navigation checks.
- CLI: search, hotel, avail, signup (and duplicate refused), book, booking, review (refused before stay), cancel, guest, menu.
- Production mode: FastAPI serves the built site, manifest, service worker and API from one origin.

Session 4 — 29 October 2026: Published on GitHub, ready for Vercel

The project is published as a public repository, github.com/A263-abdul/yupreserved, in eight commits that follow the folder structure, with an MIT licence and topic tags. It can now also be deployed on Vercel: the website as static files and the API as one Python serverless function.

Decisions and why

One commit per layer. Commits follow the architecture: setup, dataset, database, backend, CLI, frontend, docs, deployment. The history then reads like the report and each layer can be reviewed on its own.

What is not in the repository. node_modules, the built site, caches, .env files and local tooling are ignored. The PostgreSQL data directory lives outside the project, so the database is rebuilt from the CSVs with scripts/db.sh build. No passwords are committed: deploy-db.sh generates fresh role passwords on the hosted database.

Licence. Code under MIT; the source dataset and the Wikimedia photos keep their own licences (credits stored per photo).

Vercel: static site + one Python function. Vercel serves the built React site from its CDN and runs the unchanged FastAPI app as a single function (api/index.py); vercel.json sends every /api/* request to it and every other path to index.html for React Router. The database stays on hosted PostgreSQL (Neon) because a serverless function keeps no data of its own. Region Singapore (sin1), next to the database, keeps each query's round trip short.

Connection pools in a serverless function. A function instance may skip the ASGI start-up hook, so the pools open when the module is imported. check_connection replaces connections the database closed while the instance was idle.

Files added or changed

file / folder	what it is
LICENSE	MIT licence
.gitignore	also ignores local tooling (pyrightconfig.json, .claude/)
api/index.py, api/requirements.txt	Vercel entry point: imports the FastAPI app from 03-backend
vercel.json, .vercelignore	build command, output folder, rewrites, function size and duration
scripts/deploy-db.sh	prints a random YUP_SECRET, stops on any restore error, sets the India time zone, restores the purge function's REVOKE FROM PUBLIC

Tables involved

DBMS concepts used

Verified

- 145 files (~34 MB, largest 9.5 MB) checked for secrets before committing: none found.
- Pushed to main; the repository is public and opens without signing in.
- The Vercel entry point, run locally: health, search, hotel page and a why-not scenario all answer.
- Database copied to Neon (PostgreSQL 18, Singapore) in 76 s: row counts of all 38 tables, every table grant and every function grant identical to the local database; 177 constraints and 8 triggers present.
- API against Neon: OTP (wrong code refused), booking YRC3G9KA then cancelled, request without a token refused, why-not, race, deadlock, plan and 5NF demos all answer.

Part B · Project report

1. Introduction and problem statement

Travellers in India book everything from a sea-facing heritage hotel in Mumbai to a ₹1,500 guest house near the ghats of Varanasi. Behind every booking a platform must answer hard questions instantly and correctly: Which rooms are free for these dates? What does each night cost during Diwali? Can two people book the last room at the same moment? **YupReserved** is a hotel reservation platform whose database answers these questions with relational guarantees instead of application code.

problem	how YupReserved solves it
Listing data is messy: lists in one cell, inconsistent state names	A cleaning pipeline normalises the real dataset into 15 tables (up to BCNF / 4NF, rooms deliberately 3NF)
A room must never be sold twice for overlapping nights	Trigger trg_booking_rooms_guard + row locks (SELECT ... FOR UPDATE) in fn_book_stay
Prices change with festivals, weekends and GST slabs	seasons table (EXCLUDE: no overlaps) + fn_quote_stay computes every night
Invalid data (check-out before check-in, bad phone numbers, under-age guests)	Domains, CHECK, UNIQUE and foreign keys enforced by the DBMS itself
Users need to know why an action was refused	The API translates the violated constraint into a plain-language rule
Reports need joins across many tables	Views (v_hotel_cards, v_booking_summary) and a materialized view

1.1 Technologies

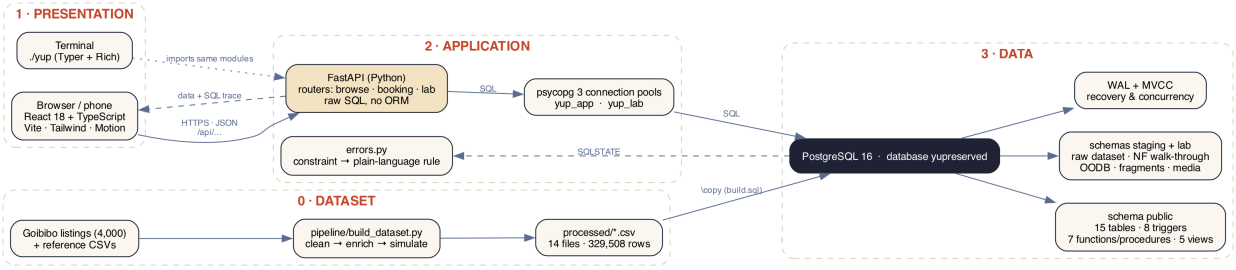
layer	technology	why
Database	PostgreSQL 16	full SQL standard support, enforced CHECK, EXCLUDE, deferred constraints, MVCC, WAL, EXPLAIN ANALYZE
Dataset pipeline	Python 3.13, pandas	repeatable, seeded cleaning and simulation
Backend	FastAPI, psycopg 3, connection pool	small, fast, typed; raw SQL so every query is visible
CLI	Typer + Rich	retrieve tables and run every demo from the terminal
Frontend	React 18, TypeScript, Vite, Tailwind CSS 4, Motion, TanStack Query	fast to build, small static bundle, deploys to any static host
Diagrams / report	Graphviz, ReportLab	diagrams and this report are generated from the live catalog

1.2 Basic concepts used

term	meaning	where in YupReserved
Relation (table)	a set of tuples with the same attributes	15 tables in schema public
Primary key	minimal attribute set that identifies each tuple; never NULL	every table, e.g. bookings(booking_id)
Candidate / alternate key	another minimal unique identifier not chosen as PK	guests.phone, guests.email, bookings.booking_ref
Foreign key	attribute(s) that must match a key of another (or the same) table	17 FKs, e.g. bookings.hotel_id → hotels
Domain	named type with a rule	indian_mobile, email_address, rupees
Trigger	event → condition → action rule inside the DBMS	8 triggers (04_triggers.sql)
View	stored query, virtual table	v_hotel_cards, v_booking_summary, v_guest_directory
Transaction (ACID)	all-or-nothing unit of work	fn_book_stay, sp_cancel_booking

2. System architecture and workflow

YupReserved is a **three-tier** system with a fourth, offline data tier that prepares the dataset. The browser never talks to the database: it calls the FastAPI application over HTTP/JSON, and only the application holds database credentials. The CLI reuses the backend's own modules, so the CLI, the API and the website run the same SQL.

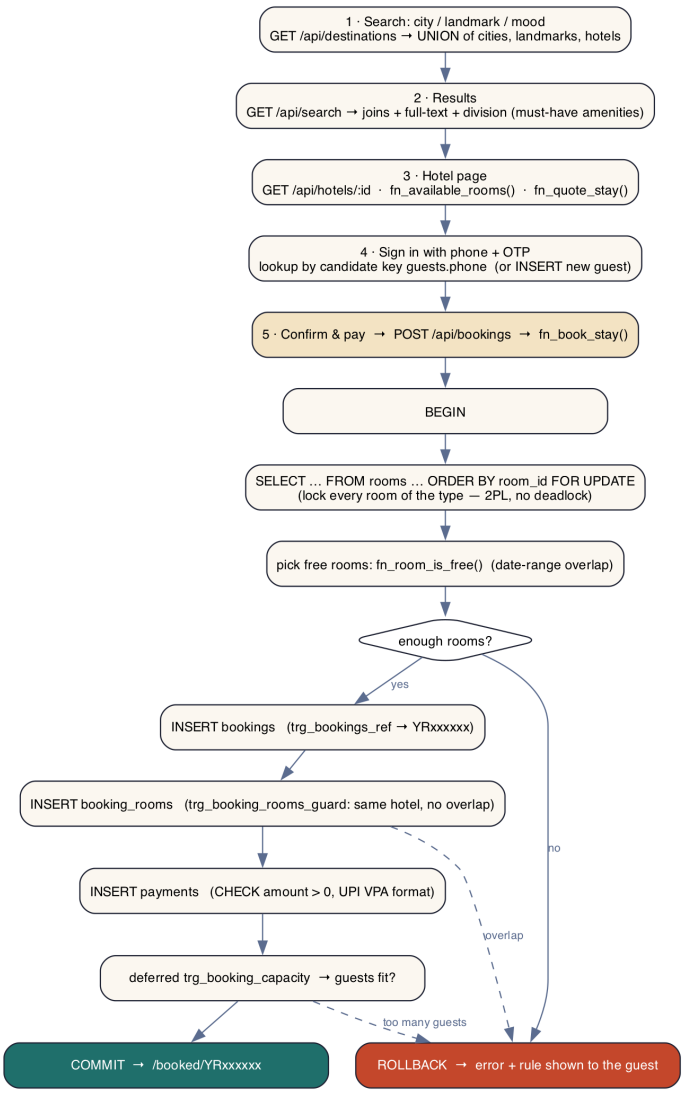


Why this architecture

- **Separation of concerns** — presentation, business logic and data can be changed independently (data independence at the application level).
- **Security** — the browser has no SQL access; the API connects as yup_app, a least-privilege role that cannot DELETE bookings or touch rooms directly.
- **Rules live in the database** — constraints and triggers protect the data whichever program writes to it (website, CLI or psql).
- **Transparency** — every API response carries the SQL it ran, which the website shows in its 'Under the hood' drawer.
- **Laptop-friendly** — PostgreSQL runs with 128 MB shared buffers and 30 connections; the frontend is static files.

2.1 Booking workflow

The workflow below follows one guest from search to a committed booking. Steps 5–9 run inside a single database transaction in the function fn_book_stay.


1 · Search: city / landmark / mood
GET /api/destinations → UNION of cities, landmarks, hotels
2 · Results
GET /api/search → joins + full-text + division (must-have amenities)
3 · Hotel page
GET /api/hotels/:id · fn_available_rooms() · fn_quote_stay()
4 · Sign in with phone + OTP
lookup by candidate key guests.phone (or INSERT new guest)
5 · Confirm & pay → POST /api/bookings → fn_book_stay()
BEGIN
SELECT … FROM rooms … ORDER BY room_id FOR UPDATE
(lock every room of the type — 2PL, no deadlock)
pick free rooms: fn_room_is_free() (date-range overlap)
enough rooms?
yes
no
INSERT bookings (trg_bookings_ref → YRxxxxxx)
INSERT booking_rooms (trg_booking_rooms_guard: same hotel, no overlap)
overlap
INSERT payments (CHECK amount > 0, UPI VPA format)
deferred trg_booking_capacity → guests fit?
too many guests
COMMIT → /booked/YRxxxxxx
ROLLBACK → error + rule shown to the guest

3. Dataset

Source: 'Indian Hotels on Goibibo' (PromptCloud, published on Kaggle; mirrored on Hugging Face as jason1966/PromptCloudHQ_hotels-on-goibibo): 4,000 listings, 36 columns, crawled in 2016. It is kept untouched in 01-dataset/raw/ and also loaded as-is into staging.goibibo_raw for the normalisation walk-through.

cleaning rule	rows
C0 raw rows	4000
C1 duplicate (name, city) removed	1
C2 coordinates outside India removed	3
C3 chain-brand listings removed	431
C4 unknown state / property type removed	0
C5 clean listings kept	3565

Other cleaning: state names normalised to 2-letter codes (e.g. 'ANDHRA PRADESH', 'Andhra Pradesh' → AP; 'Orissa' → OD); 2016-era filings corrected (Hyderabad → Telangana, Leh → Ladakh, Chandigarh → its own UT); 'Cochin' / 'Ernakulam' → Kochi; 356 raw facility labels mapped to 52 canonical amenities (processed/_amenity_mapping.csv); check-in/out times parsed from the policy text; HTML entities decoded.

Enrichment and simulation. The listings give one room type and a room count per hotel, but no prices, guests or bookings. The pipeline (seed 2026, so it is reproducible) adds room-type tiers, physical rooms, a festival calendar, 26,000 guests, 72,000 bookings and 19,041 reviews. Base prices follow a documented model (star rating × property type × city factor × review quality, with noise); nightly prices and all payments are then computed inside PostgreSQL. Resulting base-price quartiles: ₹1,699 / ₹2,699 / ₹5,299. Eighteen 'Signature' hotels are original YupReserved properties written for the platform.

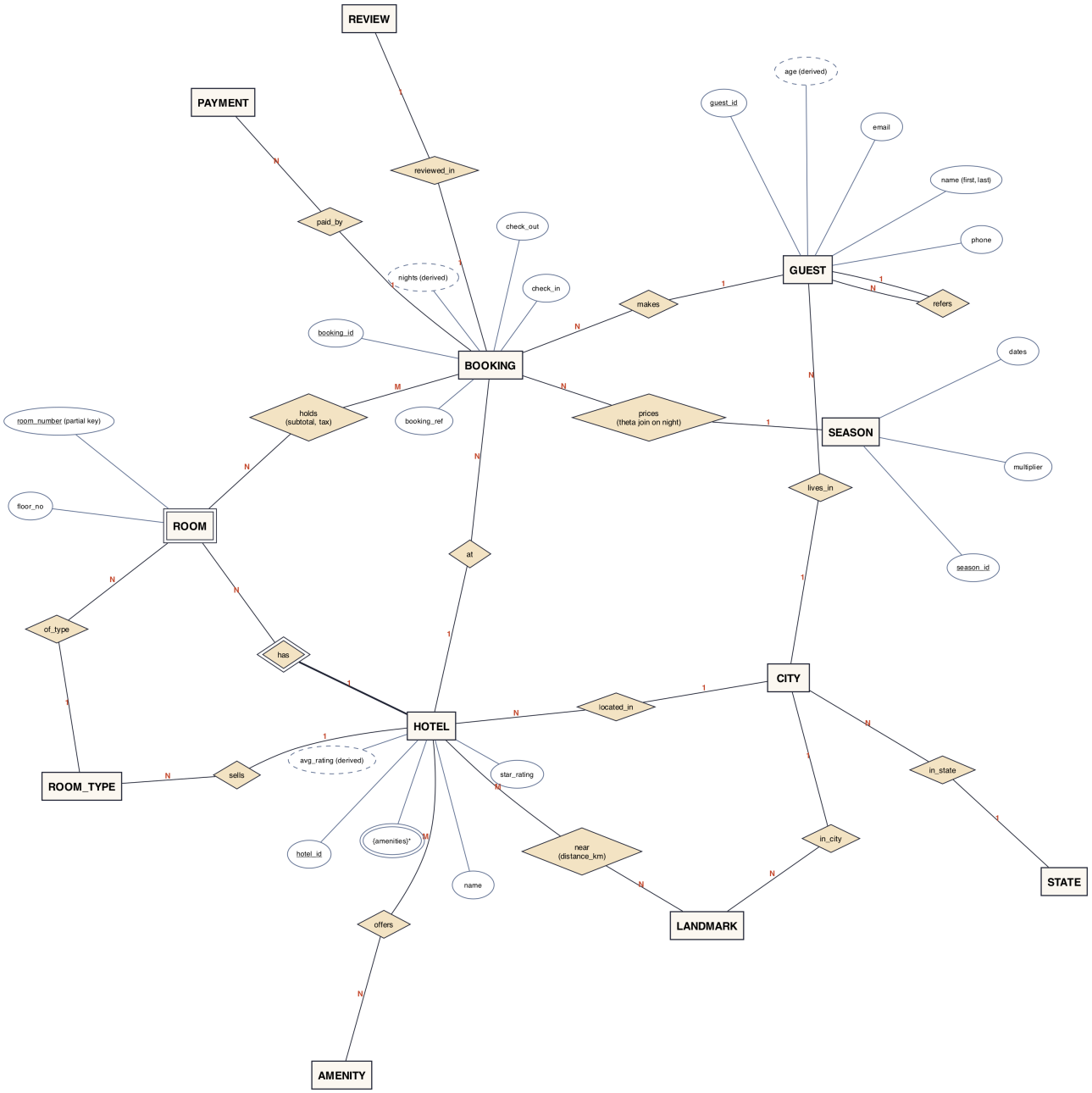
3.1 Rows per table

table_name	rows
seasons	16
states	36
amenities	52
landmarks	2056
cities	543
hotels	3583
room_types	8626
hotel_landmarks	17556
reviews	19041
guests	26002
hotel_amenities	38107
rooms	58511
payments	78688
bookings	72002
booking_rooms	83383

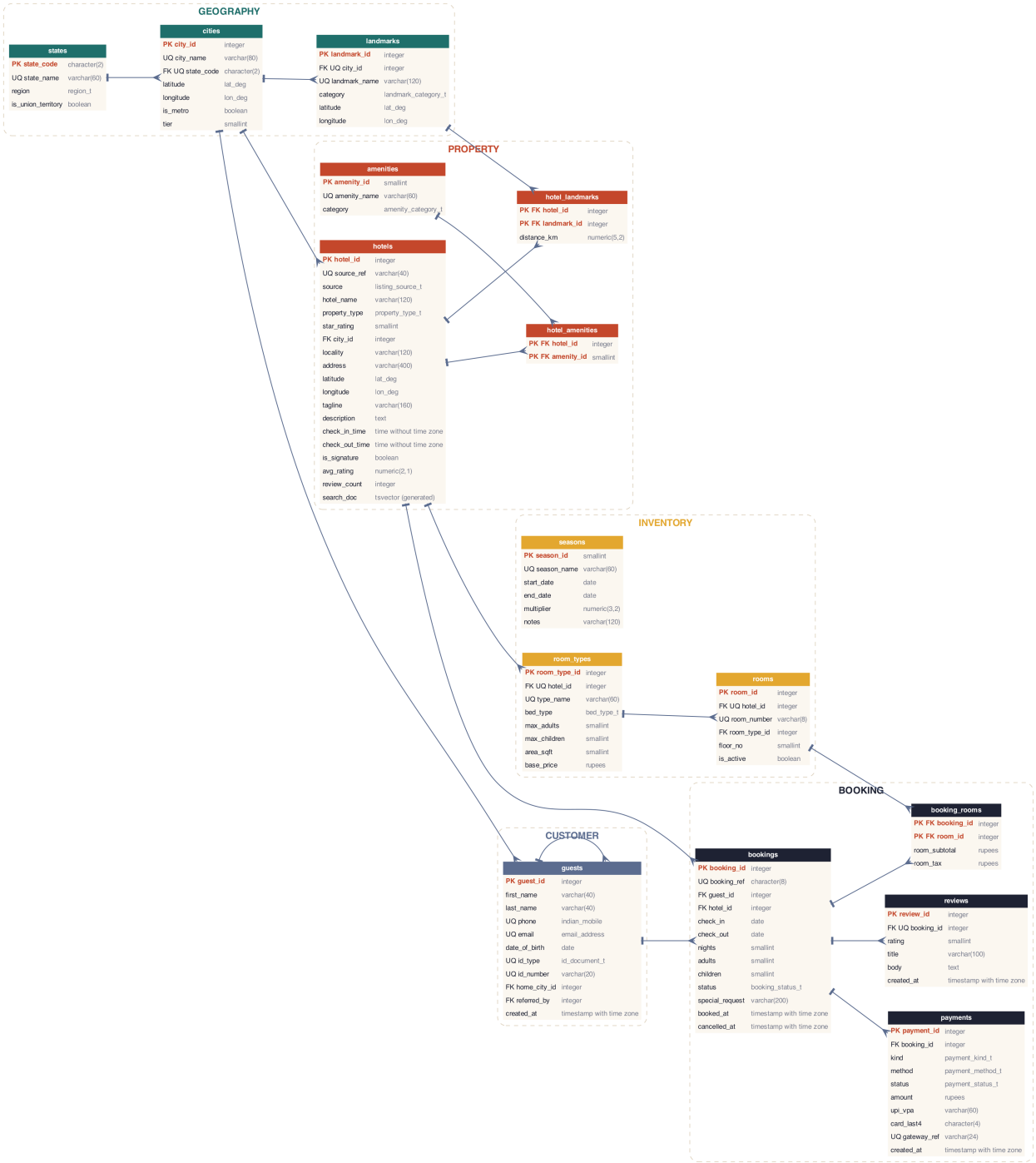
4. Database design

4.1 Conceptual model — ER diagram (Chen notation)

Rectangles are entities, diamonds relationships, ellipses attributes. ROOM is a **weak entity** (double box) identified through HOTEL by the partial key room_number; 'has' is its identifying relationship. {amenities} is a multivalued attribute (double ellipse), dashed ellipses are derived attributes, and 'refers' is a recursive relationship on GUEST.



4.2 Logical model — relational schema (generated from the live catalog)



4.3 ER → relational mapping

ER construct	mapped to	example
Strong entity	own table, own PK	HOTEL → hotels(hotel_id)
Weak entity + identifying relationship	table with owner's key + partial key	ROOM → rooms: candidate key (hotel_id, room_number); surrogate room_id as PK
1:N relationship	FK on the N side	located_in → hotels.city_id
M:N relationship	separate table, composite PK of both keys	offers → hotel_amenities(hotel_id, amenity_id)
M:N with attributes	attributes go into the relationship table	near(distance_km) → hotel_landmarks.distance_km; holds(subtotal, tax) → booking_rooms
1:1 relationship	FK + UNIQUE	reviewed_in → reviews.booking_id UNIQUE
Recursive relationship	self-referencing FK	refers → guests.referred_by → guests.guest_id
Multivalued attribute	separate table (1NF)	{amenities} → hotel_amenities
Composite attribute	one column per component	name → first_name, last_name
Derived attribute	computed, or stored and maintained	nights GENERATED; age in a view; avg_rating kept by trigger

4.4 The 15 tables

#	table	stores	primary key	references
1	states	Indian states and union territories (natural key)	state_code	—
2	cities	cities, each in one state	city_id	states
3	landmarks	places guests search by	landmark_id	cities
4	hotels	one row per property	hotel_id	cities
5	hotel_landmarks	which landmarks are near which hotel, with distance	hotel_id, landmark_id	hotels, landmarks
6	amenities	controlled amenity vocabulary	amenity_id	—
7	hotel_amenities	which hotel offers which amenity	hotel_id, amenity_id	amenities, hotels
8	room_types	what a hotel sells, with list price	room_type_id	hotels
9	rooms	physical rooms (weak entity of hotel)	room_id	room_types
10	seasons	festival / holiday pricing calendar	season_id	—
11	guests	customers	guest_id	cities, guests
12	bookings	reservation header: guest, hotel, dates, status	booking_id	guests, hotels
13	booking_rooms	rooms held by a booking, with agreed price	booking_id, room_id	bookings, rooms
14	payments	charges and refunds	payment_id	bookings
15	reviews	one review per completed stay	review_id	bookings

5. Keys

Super key: any attribute set that uniquely identifies a tuple (e.g. {booking_id, guest_id}). **Candidate key**: a minimal super key. **Primary key**: the candidate key chosen to identify rows; the others become **alternate keys** enforced with UNIQUE. A **surrogate** key is a system-generated number with no business meaning; a **natural** key comes from the real world.

Primary keys of all 15 tables [catalog.primary_keys]

RA / RC: — (catalog)

11 single-column surrogate/natural keys and 4 composite keys (on the M:N relationship tables).

```
SELECT con.conrelid::regclass::text AS table_name, con.conname AS constraint_name,
       string_agg(a.attname, ', ' ORDER BY k.ord) AS key_columns,
       CASE WHEN count(*) > 1 THEN 'composite' ELSE 'single' END AS kind
FROM pg_constraint con
JOIN pg_namespace n ON n.oid = con.connamespace
CROSS JOIN LATERAL unnest(con.conkey) WITH ORDINALITY AS k(attnum, ord)
JOIN pg_attribute a ON a.attrelid = con.conrelid AND a.attnum = k.attnum
WHERE con.contype = 'p' AND n.nspname = 'public'
GROUP BY con.conrelid, con.conname
ORDER BY kind DESC, table_name
```

table_name	constraint_name	key_columns	kind
amenities	pk_amenities	amenity_id	single
bookings	pk_bookings	booking_id	single
cities	pk_cities	city_id	single
guests	pk_guests	guest_id	single
hotels	pk_hotels	hotel_id	single
landmarks	pk_landmarks	landmark_id	single
payments	pk_payments	payment_id	single
reviews	pk_reviews	review_id	single
room_types	pk_room_types	room_type_id	single
rooms	pk_rooms	room_id	single
seasons	pk_seasons	season_id	single
states	pk_states	state_code	single
booking_rooms	pk_booking_rooms	booking_id, room_id	composite
hotel_amenities	pk_hotel_amenities	hotel_id, amenity_id	composite
hotel_landmarks	pk_hotel_landmarks	hotel_id, landmark_id	composite

table	why this primary key
states (state_code)	natural key: 2-letter codes are short, stable and universally used (MH, RJ), so no surrogate is needed
cities, landmarks, hotels, room_types, guests, bookings, payments, reviews (…_id)	surrogate integer identity: natural candidates (names, phone, email) can change or be long; a small integer makes joins and FKs fast and …
rooms (room_id)	surrogate PK; the natural candidate key (hotel_id, room_number) is kept as UNIQUE. A single-column key keeps booking_rooms in 2NF
amenities, seasons (smallint)	small lookup tables: 2-byte keys are enough
hotel_amenities, hotel_landmarks, booking_rooms (composite)	M:N relationship tables: the pair of foreign keys IS the identity; a pair can appear only once

Alternate (candidate) keys — every UNIQUE constraint [catalog.candidate_keys]

RA / RC: — (catalog)

A candidate key that was not chosen as the primary key is an alternate key, enforced with UNIQUE.

```
SELECT con.conrelid::regclass AS table_name, con.conname AS constraint_name,
```

```
       pg_get_constraintdef(con.oid) AS definition
FROM pg_constraint con JOIN pg_namespace n ON n.oid = con.connamespace
WHERE con.contype = 'u' AND n.nspname = 'public'
ORDER BY 1, 2
```

table_name	constraint_name	definition
states	uq_states_name	UNIQUE (state_name)
cities	uq_cities_name_state	UNIQUE (city_name, state_code)
landmarks	uq_landmarks_city_name	UNIQUE (city_id, landmark_name)
hotels	uq_hotels_source_ref	UNIQUE (source_ref)
amenities	uq_amenities_name	UNIQUE (amenity_name)
room_types	uq_room_types_hotel_name	UNIQUE (hotel_id, type_name)
room_types	uq_room_types_id_hotel	UNIQUE (room_type_id, hotel_id)
rooms	uq_rooms_hotel_number	UNIQUE (hotel_id, room_number)
seasons	uq_seasons_name	UNIQUE (season_name)
guests	uq_guests_email	UNIQUE (email)
guests	uq_guests_id_document	UNIQUE (id_type, id_number)
guests	uq_guests_phone	UNIQUE (phone)
bookings	uq_bookings_ref	UNIQUE (booking_ref)
payments	uq_payments_gateway_ref	UNIQUE (gateway_ref)
reviews	uq_reviews_booking	UNIQUE (booking_id)

Foreign keys — child → parent, with ON DELETE rule [catalog.foreign_keys]

RA / RC: — (catalog)

Every arrow of the ER diagram is one row here. ON DELETE says what happens to children when a parent row is deleted.

```
SELECT con.conrelid::regclass AS child_table,
       pg_get_constraintdef(con.oid) AS definition,
       con.conname AS constraint_name,
       CASE con.confdeltype WHEN 'a' THEN 'NO ACTION' WHEN 'r' THEN 'RESTRICT'
            WHEN 'c' THEN 'CASCADE' WHEN 'n' THEN 'SET NULL' WHEN 'd' THEN 'SET DEFAULT' END AS on_delete
FROM pg_constraint con
JOIN pg_namespace n ON n.oid = con.connamespace
WHERE con.contype = 'f' AND n.nspname = 'public'
ORDER BY 1, 3
```

child_table	definition	constraint_name	on_delete
cities	FOREIGN KEY (state_code) REFERENCES states(state_code) ON UPDATE CASCADE ON DELETE RESTRICT	fk_cities_states	RESTRICT
landmarks	FOREIGN KEY (city_id) REFERENCES cities(city_id) ON DELETE CASCADE	fk_landmarks_cities	CASCADE
hotels	FOREIGN KEY (city_id) REFERENCES cities(city_id) ON DELETE RESTRICT	fk_hotels_cities	RESTRICT
hotel_landmarks	FOREIGN KEY (hotel_id) REFERENCES hotels(hotel_id) ON DELETE CASCADE	fk_hotel_landmarks_hotels	CASCADE
hotel_landmarks	FOREIGN KEY (landmark_id) REFERENCES landmarks(landmark_id) ON DELETE CASCADE	fk_hotel_landmarks_landmarks	CASCADE
hotel_amenities	FOREIGN KEY (amenity_id) REFERENCES amenities(amenity_id) ON DELETE RESTRICT	fk_hotel_amenities_amenities	RESTRICT
hotel_amenities	FOREIGN KEY (hotel_id) REFERENCES hotels(hotel_id) ON DELETE CASCADE	fk_hotel_amenities_hotels	CASCADE
room_types	FOREIGN KEY (hotel_id) REFERENCES hotels(hotel_id) ON DELETE CASCADE	fk_room_types_hotels	CASCADE
rooms	FOREIGN KEY (room_type_id, hotel_id) REFERENCES room_types(room_type_id, hotel_id) ON DELETE CASCADE	fk_rooms_room_types	CASCADE

child_table	definition	constraint_name	on_delete
guests	FOREIGN KEY (home_city_id) REFERENCES cities(city_id) ON DELETE SET NULL	fk_guests_cities	SET NULL
guests	FOREIGN KEY (referred_by) REFERENCES guests(guest_id) ON DELETE SET NULL	fk_guests_referrer	SET NULL
bookings	FOREIGN KEY (guest_id) REFERENCES guests(guest_id) ON DELETE RESTRICT	fk_bookings_guests	RESTRICT
bookings	FOREIGN KEY (hotel_id) REFERENCES hotels(hotel_id) ON DELETE RESTRICT	fk_bookings_hotels	RESTRICT
booking_rooms	FOREIGN KEY (booking_id) REFERENCES bookings(booking_id) ON DELETE CASCADE	fk_booking_rooms_bookings	CASCADE
booking_rooms	FOREIGN KEY (room_id) REFERENCES rooms(room_id) ON DELETE RESTRICT	fk_booking_rooms_rooms	RESTRICT
payments	FOREIGN KEY (booking_id) REFERENCES bookings(booking_id) ON DELETE RESTRICT	fk_payments_bookings	RESTRICT
reviews	FOREIGN KEY (booking_id) REFERENCES bookings(booking_id) ON DELETE CASCADE	fk_reviews_bookings	CASCADE

Why these ON DELETE rules

- **RESTRICT** where history matters: a hotel, guest or booking that is still referenced cannot disappear (bookings → hotels, payments → bookings).
- **CASCADE** for composition: a hotel's room types, rooms, amenity links and landmark links are part of the hotel; a booking's booking_rooms are part of the booking.
- **SET NULL** for optional links: if a referrer guest or a home city is removed, the guest stays and the link becomes unknown.
- **Composite FK** rooms(room_type_id, hotel_id) → room_types(room_type_id, hotel_id): a room can never point at another hotel's room type.

6. Constraints, triggers, functions and views

6.1 Domains

domain	rule
rupees	numeric(12,2), ≥ 0
indian_mobile	10 digits, first digit 6–9
email_address	regular expression for an email
lat_deg / lon_deg	valid latitude / longitude

CHECK and EXCLUDE constraints [catalog.checks]

RA / RC: — (catalog)

Rules the database enforces on every INSERT / UPDATE, whatever program sends it.

```
SELECT con.conrelid::regclass AS table_name, con.conname AS constraint_name,
       CASE con.contype WHEN 'c' THEN 'CHECK' WHEN 'x' THEN 'EXCLUDE' END AS kind,
       pg_get_constraintdef(con.oid) AS rule
FROM pg_constraint con JOIN pg_namespace n ON n.oid = con.connamespace
WHERE con.contype IN ('c', 'x') AND n.nspname = 'public' AND con.conrelid <> 0
ORDER BY 1, 2
```

table_name	constraint_name	kind	rule
states	ck_states_code_format	CHECK	CHECK ((state_code ~ '^[A-Z]{2}$'::text))
cities	ck_cities_metro_tier	CHECK	CHECK (((NOT is_metro) OR (tier = 1)))
cities	ck_cities_photo	CHECK	CHECK (((photo_url IS NULL) OR ((photo_url)::text ~ '^https://'::text) AND (photo_credit IS NOT NULL))))
cities	ck_cities_tier	CHECK	CHECK (((tier >= 1) AND (tier <= 3)))
landmarks	ck_landmarks_coords	CHECK	CHECK (((latitude IS NULL) = (longitude IS NULL)))
landmarks	ck_landmarks_photo	CHECK	CHECK (((photo_url IS NULL) OR ((photo_url)::text ~ '^https://'::text) AND (photo_credit IS NOT NULL))))
hotels	ck_hotels_in_india	CHECK	CHECK (((((latitude)::numeric >= (6)::numeric) AND ((latitude)::numeric <= 37.5)) AND (((longitude)::numeric >= (68)::numeric) AND ((long…
hotels	ck_hotels_rating_count	CHECK	CHECK (((review_count >= 0) AND ((avg_rating IS NULL) = (review_count = 0))))
hotels	ck_hotels_rating_range	CHECK	CHECK (((avg_rating >= (1)::numeric) AND (avg_rating <= (5)::numeric)))
hotels	ck_hotels_signature	CHECK	CHECK ((is_signature = (source = 'YUPRESERVED'::listing_source_t)))
hotels	ck_hotels_star	CHECK	CHECK (((star_rating >= 0) AND (star_rating <= 5)))
hotel_landmarks	ck_hotel_landmarks_distance	CHECK	CHECK (((distance_km >= (0)::numeric) AND (distance_km < (100)::numeric)))
room_types	ck_room_types_adults	CHECK	CHECK (((max_adults >= 1) AND (max_adults <= 6)))
room_types	ck_room_types_area	CHECK	CHECK (((area_sqft >= 50) AND (area_sqft <= 5000)))
room_types	ck_room_types_children	CHECK	CHECK (((max_children >= 0) AND (max_children <= 4)))
room_types	ck_room_types_price	CHECK	CHECK ((((base_price)::numeric >= (299)::numeric) AND ((base_price)::numeric <= (500000)::numeric)))
rooms	ck_rooms_floor	CHECK	CHECK (((floor_no >= 0) AND (floor_no <= 60)))
seasons	ck_seasons_dates	CHECK	CHECK ((end_date >= start_date))
seasons	ck_seasons_multiplier	CHECK	CHECK (((multiplier >= 0.50) AND (multiplier <= 3.00)))
seasons	ex_seasons_no_overlap	EXCLUDE	EXCLUDE USING gist (daterange(start_date, end_date, '[]'::text) WITH &&)
guests	ck_guests_adult	CHECK	CHECK ((date_of_birth <= (((created_at AT TIME ZONE 'Asia/Kolkata'::text))::date - '18 years'::interval)))

table_name	constraint_name	kind	rule
guests	ck_guests_not_self	CHECK	CHECK ((referred_by <> guest_id))
bookings	ck_bookings_booked_before_arrival	CHECK	CHECK ((((booked_at AT TIME ZONE 'Asia/Kolkata'::text))::date <= check_in))
bookings	ck_bookings_cancel_after_booking	CHECK	CHECK ((cancelled_at >= booked_at))
bookings	ck_bookings_cancel_state	CHECK	CHECK (((status = 'CANCELLED'::booking_status_t) = (cancelled_at IS NOT NULL)))
bookings	ck_bookings_dates	CHECK	CHECK ((check_out > check_in))
bookings	ck_bookings_guests	CHECK	CHECK (((adults >= 1) AND (children >= 0) AND ((adults + children) <= 24)))
bookings	ck_bookings_max_stay	CHECK	CHECK (((check_out - check_in) <= 30))
bookings	ck_bookings_ref_format	CHECK	CHECK ((booking_ref ~ '^YR[A-Z0-9]{6}$'::text))
booking_rooms	ck_booking_rooms_amount	CHECK	CHECK (((room_subtotal)::numeric > (0)::numeric))
payments	ck_payments_amount	CHECK	CHECK (((amount)::numeric > (0)::numeric))
payments	ck_payments_card	CHECK	CHECK (((card_last4 IS NULL) OR ((method = 'CARD'::payment_method_t) AND (card_last4 ~ '^[0-9]{4}$'::text))))
payments	ck_payments_upi	CHECK	CHECK (((upi_vpa IS NULL) OR ((method = 'UPI'::payment_method_t) AND ((upi_vpa)::text ~ '^[a-z0-9._-]{2,}@[a-z]{2,}$'::text))))
payments	ck_payments_upi_vpa_required	CHECK	CHECK (((method <> 'UPI'::payment_method_t) OR (upi_vpa IS NOT NULL)))
reviews	ck_reviews_rating	CHECK	CHECK (((rating >= 1) AND (rating <= 5)))

Triggers [catalog.triggers]

RA / RC: — (catalog)

Event → condition → action rules that run inside the database. A constraint trigger can be DEFERRED to COMMIT time.

```
SELECT t.tgrelid::regclass::text AS table_name, t.tgname AS trigger_name,
       CASE WHEN t.tgconstraint <> 0 THEN 'constraint trigger' ELSE 'trigger' END AS kind,
       substring(pg_get_triggerdef(t.oid) FROM '(BEFORE|AFTER) .* ON') AS fires,
       t.tgfoid::regproc AS function_name
FROM pg_trigger t
JOIN pg_class c ON c.oid = t.tgrelid JOIN pg_namespace n ON n.oid = c.relnamespace
WHERE NOT t.tgisinternal AND n.nspname = 'public'
ORDER BY 1, 2
```

table_name	trigger_name	kind	fires	function_name
booking_rooms	trg_booking_capacity	constraint trigger	AFTER	tg_booking_capacity
booking_rooms	trg_booking_rooms_guard	trigger	BEFORE	tg_booking_rooms_guard
bookings	trg_bookings_dates	trigger	BEFORE	tg_bookings_dates
bookings	trg_bookings_ref	trigger	BEFORE	tg_bookings_ref
bookings	trg_bookings_status	trigger	BEFORE	tg_bookings_status
payments	trg_payments_refund_guard	trigger	BEFORE	tg_payments_refund_guard
reviews	trg_reviews_guard	trigger	BEFORE	tg_reviews_guard
reviews	trg_reviews_rollup	trigger	AFTER	tg_reviews_rollup

6.2 Functions and procedures

routine	type	what it does
fn_gst_rate(tariff)	function (IMMUTABLE)	GST slab: 0% ≤ ₹1,000 · 5% ≤ ₹7,500 · 18% above

routine	type	what it does
fn_quote_stay(type, in, out, rooms)	set-returning function	night-by-night price: base × season × weekend, then GST
fn_stay_total(…)	function	sum of fn_quote_stay
fn_room_is_free(room, in, out)	function	overlap test on half-open date ranges
fn_available_rooms(hotel, in, out)	set-returning function	free rooms per room type
fn_book_stay(…)	function, SECURITY DEFINER	the whole booking as one transaction with row locks
sp_cancel_booking(id, OUT refund)	PROCEDURE (CALL)	cancel + refund policy (100% / 50% / 0%)

6.3 Views

view	purpose
v_hotel_cards	hotel + city + state + cheapest price (search cards)
v_booking_summary	booking + guest + hotel + rooms + money (front desk, My trips)
v_guest_directory	security view: masked phone/email; the only guest data the analyst role can read
v_upcoming_arrivals	updatable view WITH CHECK OPTION: edits cannot push a row out of the view
mv_city_performance	materialized view: revenue, room nights, ADR per city (REFRESH to update)

7. Functional dependencies and normalization

A functional dependency X → Y means: any two tuples that agree on X also agree on Y. Normalisation removes dependencies that cause redundancy and update, insertion and deletion anomalies. YupReserved was normalised **on the real raw dataset**; every step below exists as tables in schema lab.

7.1 Functional dependencies of the final design

table	functional dependencies	highest NF
states	state_code → state_name, region, is_ut; state_name → state_code	BCNF
cities	city_id → city_name, state_code, lat, lon, is_metro, tier; (city_name, state_code) → city_id	BCNF
landmarks	landmark_id → city_id, name, category, lat, lon; (city_id, name) → landmark_id	BCNF
hotels	hotel_id → all attributes; source_ref → hotel_id	BCNF
hotel_landmarks	(hotel_id, landmark_id) → distance_km	BCNF / 4NF
amenities	amenity_id → name, category; name → amenity_id	BCNF
hotel_amenities	no non-trivial FD (all-key)	BCNF / 4NF
room_types	room_type_id → hotel_id, type_name, bed, capacity, area, base_price; (hotel_id, type_name) → room_type_id	BCNF
rooms	room_id → hotel_id, room_number, room_type_id; (hotel_id, room_number) → room_id; room_type_id → hotel_id	3NF (not BCNF, by choice)
seasons	season_id → name, dates, multiplier; name → season_id	BCNF
guests	guest_id → all; phone → guest_id; email → guest_id; (id_type, id_number) → guest_id	BCNF
bookings	booking_id → guest, hotel, dates, guests, status, …; booking_ref → booking_id; (check_in, check_out) → nights	3NF*
booking_rooms	(booking_id, room_id) → room_subtotal, room_tax	BCNF
payments	payment_id → booking_id, kind, method, status, amount, …; gateway_ref → payment_id	BCNF
reviews	review_id → booking_id, rating, title, body, created_at; booking_id → review_id	BCNF

* nights is a **generated column**: the DBMS computes it from (check_in, check_out), so it can never disagree with them. The redundancy is controlled, not an anomaly.

7.2 UNF → 1NF → 2NF → 3NF on the raw data

UNF — the raw listing stores facilities as a list in one cell [nf.unf]

RA / RC: —

hotel_facilities holds a repeating group ("A|B|C|..."): not atomic, so not in 1NF. You cannot index, count or join on one facility.

```
SELECT property_id, property_name, city, left(hotel_facilities, 120) || '…' AS hotel_facilities
FROM lab.unf_listing
WHERE hotel_facilities IS NOT NULL
LIMIT 5
```

property_id	property_name	city	hotel_facilities
HTLGBO1000016238	Baragarh Regency	Manali	Doctor on Call|Dry Cleaning|Laundry Service Available|Lobby|Parking Facilities Available|Gardens|Dance Performances (on …
HTLGBO1000015822	Asian Suites A- 585	Gurgaon	Airport Transfer Available / Surcharge|Banquet Facilities|Business Center |Currency Exchange|Doctor on Call|Health Club …

property_id	property_name	city	hotel_facilities
HTLGBO10000004551	Bevvan Resort	Goa	Swimming Pool|Bar / Lounge |Laundry Service Available|Multi Cuisine Restaurant...

... 2 more row(s) not shown.

FD violated in the RAW data — city name → state does NOT hold

[fd.violation_raw]

RA / RC: σ count(DISTINCT state) > 1 (city γ ... (staging.goibibo_raw))

In the raw dataset the same city name appears under different state spellings, so 'city' alone cannot determine 'state'. That is why cities has key city_id and candidate key (city_name, state_code).

```
SELECT city, count(DISTINCT state) AS different_state_values, string_agg(DISTINCT state, ' | ') AS values_seen
FROM staging.goibibo_raw
GROUP BY city
HAVING count(DISTINCT state) > 1
ORDER BY 2 DESC, 1
```

city	different_state_values	values_seen
Aurangabad	2	Bihar | Maharashtra
Bangalore	2	KARNATAKA | Karnataka
Chennai	2	TAMIL NADU | Tamil Nadu
Delhi	2	DELHI | Delhi
Ernakulam	2	KERALA | Kerala
Goa	2	GOA | Goa
Hyderabad	2	ANDHRA PRADESH | Andhra Pradesh
Mumbai	2	MAHARASHTRA | Maharashtra

Why this matters. A decomposition is only lossless if the dependency it splits on really holds. When the walk-through was first run on the uncleaned names, the 3NF re-join produced 357 lost or spurious rows, because 'city → state' is false in the raw file (e.g. Hyderabad filed under both Andhra Pradesh and Telangana). City names are not even unique after cleaning — Aurangabad exists in both Maharashtra and Bihar. So the walk-through (and the real cities table) uses the surrogate city_id, for which city_id → city_name, state holds.

1NF — one atomic facility per row (but name and city now repeat) [nf.first]

RA / RC: key = (property_id, facility)

Every cell is atomic. But property_id → property_name, city_id, city, state is a PARTIAL dependency on the key: the name and city repeat once per facility (update anomaly).

```
SELECT * FROM lab.nf1_listing_facility
WHERE property_id = (SELECT property_id FROM lab.nf1_listing_facility LIMIT 1)
ORDER BY facility
```

operty_id	property_name	city_id	city	state	facility
LGBO10000057	Hotel Samdariya inn	303	Jabalpur	Madhya Pradesh	24 Hour Front Desk
LGBO10000057	Hotel Samdariya inn	303	Jabalpur	Madhya Pradesh	ATM / Banking
LGBO10000057	Hotel Samdariya inn	303	Jabalpur	Madhya Pradesh	Air / Rail Booking
LGBO10000057	Hotel Samdariya inn	303	Jabalpur	Madhya Pradesh	Air-Conditioning - Central
LGBO10000057	Hotel Samdariya inn	303	Jabalpur	Madhya Pradesh	Baggage Room
LGBO10000057	Hotel Samdariya inn	303	Jabalpur	Madhya Pradesh	Banquet Facilities

… 42 more row(s) not shown.

2NF — split off the partial dependency [nf.second]

RA / RC: R → R1(property_id, name, city, state), R2(property_id, facility)

Now every non-key attribute depends on the WHOLE key. But property_id → city_id → city, state is TRANSITIVE: the city and state are repeated for every hotel of that city.

```
SELECT city_id, city, state, count(*) AS hotels_repeating_this_city
FROM lab.nf2_property
GROUP BY city_id, city, state
ORDER BY 3 DESC
LIMIT 10
```

city_id	city	state	hotels_repeating_this_city
526	Durgapur	West Bengal	2
531	Kalimpong	West Bengal	10
523	Dayapur	West Bengal	1
533	Kolkata	West Bengal	39
530	Jalpaiguri	West Bengal	4
543	Tajpur	West Bengal	1

… 4 more row(s) not shown.

3NF — city_id → city, state moved to its own table; lossless join check

[nf.third]

RA / RC: R1 ⋈ R2 ⋈ R3 = original 1NF relation

Joining the 3NF tables back gives exactly the 1NF rows — no spurious tuples, nothing lost. That proves the decomposition is lossless.

```
SELECT (SELECT count(*) FROM lab.nf1_listing_facility) AS rows_in_1nf,
       (SELECT count(*) FROM lab.nf3_property p
          JOIN lab.nf3_city c USING (city_id)
          JOIN lab.nf2_property_facility f USING (property_id)) AS rows_after_rejoin,
       (SELECT count(*) FROM (
            SELECT property_id, property_name, city_id, city, state, facility FROM lab.nf1_listing_facility
            EXCEPT
            SELECT p.property_id, p.property_name, p.city_id, c.city, c.state, f.facility
            FROM lab.nf3_property p JOIN lab.nf3_city c USING (city_id)
            JOIN lab.nf2_property_facility f USING (property_id)) d) AS rows_lost_or_spurious
```

rows_in_1nf	rows_after_rejoin	rows_lost_or_spurious
57088	57088	0

7.3 BCNF — and why rooms stays in 3NF

3NF but not BCNF — rooms (a deliberate choice) [nf.bcnf_rooms]

RA / RC: rooms(room_id, hotel_id, room_number, room_type_id): room_type_id → hotel_id

room_type_id → hotel_id has a determinant that is not a superkey (violates BCNF), but hotel_id is prime (part of candidate key {hotel_id, room_number}), so 3NF holds. Decomposing to BCNF would lose (hotel_id, room_number) → room_id, so we keep 3NF and enforce consistency with a composite FK.

```
SELECT r.room_type_id, count(DISTINCT r.hotel_id) AS hotels_per_room_type
FROM rooms r
GROUP BY r.room_type_id
HAVING count(DISTINCT r.hotel_id) > 1
```

room_type_id	hotels_per_room_type

In rooms, room_type_id → hotel_id has a determinant that is not a superkey, so rooms is not in BCNF. It is in 3NF because hotel_id is a **prime attribute** (part of the candidate key {hotel_id, room_number}). The BCNF decomposition R1(room_type_id, hotel_id), R2(room_id,

room_number, room_type_id) is lossless but **not dependency-preserving**: '(hotel_id, room_number) is unique' could no longer be checked in a single table. We therefore keep 3NF and guarantee consistency with the composite foreign key (room_type_id, hotel_id).

7.4 4NF and 5NF

4NF — two independent multivalued facts blow up into a cross product

[nf.fourth]

RA / RC: hotel ↠ amenity | landmark

Storing amenities and landmarks in one table forces every amenity to pair with every landmark. 4NF keeps them in two tables (hotel_amenities, hotel_landmarks).

```
SELECT hotel_name,
       count(*) AS rows_if_combined,
       count(DISTINCT amenity_name) AS amenities,
       count(DISTINCT landmark_name) AS landmarks,
       count(DISTINCT amenity_name) + count(DISTINCT landmark_name) AS rows_in_4nf
FROM lab.mvd_hotel_amenity_landmark
GROUP BY hotel_name
ORDER BY rows_if_combined DESC
```

hotel_name	rows_if_combined	amenities	landmarks	rows_in_4nf
Samudra Mahal	66	11	6	17
Lutyens Lodge	64	8	8	16
Pichola Rajbagh	40	10	4	14
Amber Haveli	40	8	5	13
Morjim Driftwood Shacks	36	6	6	12
Charminar Courtyard	28	7	4	11
Cubbon Leaf	24	6	4	10
Hooghly Ghat House	18	6	3	9

… 10 more row(s) not shown.

5NF — a join dependency: 3-way join is lossless, any 2-way join is not

[nf.fifth]

RA / RC: * [(hotel, plan), (hotel, channel), (plan, channel)]

Joining only two projections invents spurious rows; joining all three gives back exactly the original 6 rows.

```
SELECT 'original'           AS relation, count(*) AS rows FROM lab.jd_offer
UNION ALL
SELECT 'hp ⋈ hc (2-way)',   count(*) FROM lab.jd_hotel_plan hp JOIN lab.jd_hotel_channel hc USING (hotel)
UNION ALL
SELECT 'hp ⋈ hc ⋈ pc (3-way)', count(*) FROM lab.jd_hotel_plan hp
       JOIN lab.jd_hotel_channel hc USING (hotel)
       JOIN lab.jd_plan_channel pc ON pc.meal_plan = hp.meal_plan AND pc.channel = hc.channel
```

relation	rows
original	6
hp ⋈ hc (2-way)	7
hp ⋈ hc ⋈ pc (3-way)	7

8. Relational algebra, calculus and SQL

Every query below is stored in 02-database/queries with its relational-algebra (or calculus) form; the outputs are produced live from the database while this report is built.

Selection (σ) — 5-star hotels in Rajasthan [basics.selection]

RA / RC: σ star_rating = 5 ∧ state_code = 'RJ' (hotels ⋈ cities)

σ keeps the rows that satisfy a predicate. WHERE is the SQL form of σ.

```
SELECT h.hotel_id, h.hotel_name, c.city_name, h.star_rating
FROM hotels h JOIN cities c ON c.city_id = h.city_id
WHERE h.star_rating = 5 AND c.state_code = 'RJ'
ORDER BY c.city_name, h.hotel_name
```

hotel_id	hotel_name	city_name	star_rating
146	Chokhi Dhani - An Ethnic Village Resort	Jaipur	5
1277	TALAI BAGH PALACE	Jaipur	5
2216	The Raj Palace	Jaipur	5
2396	The Tree House	Jaipur	5
122	BRYS Fort	Jaisalmer	5
1264	Sujan-The Serai	Jaisalmer	5
3	Pichola Rajbagh	Udaipur	5

Projection (π) — the distinct property types on the platform

[basics.projection]

RA / RC: π property_type (hotels)

π keeps chosen columns. Relational algebra works on sets, so duplicates disappear — SQL needs DISTINCT for that.

```
SELECT DISTINCT property_type
FROM hotels
ORDER BY property_type
```

property_type
Hotel
Resort
Guest House
Homestay
Service Apartment
Lodge
Houseboat
Cottage

… 13 more row(s) not shown.

Inner (equi) join — hotels with their city and state [joins.inner]

RA / RC: π hotel_name, city_name, state_name (hotels ⋈ hotels.city_id = cities.city_id cities ⋈ cities.state_code = states.state_code states)

Keeps only pairs of rows whose join columns are equal. Every hotel has a city (FK, NOT NULL), so no hotel is lost.

```
SELECT h.hotel_name, c.city_name, s.state_name, s.region
FROM hotels h
INNER JOIN cities c ON c.city_id = h.city_id
INNER JOIN states s ON s.state_code = c.state_code
WHERE h.is_signature
ORDER BY s.region, c.city_name
```

hotel_name	city_name	state_name	region
Lutyens Lodge	Delhi	Delhi	NORTH
Solang Snowline Camp	Manali	Himachal Pradesh	NORTH
Ganga Kinara Retreat	Rishikesh	Uttarakhand	NORTH
Pine Needle Cottage	Shimla	Himachal Pradesh	NORTH
Dal Shikara House	Srinagar	Jammu and Kashmir	NORTH
Ghat Ninety-One	Varanasi	Uttar Pradesh	NORTH
Vembanad Kettuvallam	Alleppey	Kerala	SOUTH
Cubbon Leaf	Bangalore	Karnataka	SOUTH

… 10 more row(s) not shown.

Natural join — cities and states on their common column [joins.natural]

RA / RC: cities * states

NATURAL JOIN joins on every column with the same name (here state_code) and shows it once. Convenient, but fragile: adding a same-named column silently changes the join.

```
SELECT city_name, state_code, state_name, region
FROM cities NATURAL JOIN states
WHERE region = 'NORTHEAST'
ORDER BY state_name, city_name
```

city_name	state_code	state_name	region
Bomdila	AR	Arunachal Pradesh	NORTHEAST
Bongaigaon	AS	Assam	NORTHEAST
Dibrugarh	AS	Assam	NORTHEAST
Golaghat	AS	Assam	NORTHEAST
Guwahati	AS	Assam	NORTHEAST
Kaziranga	AS	Assam	NORTHEAST
Sibsagar	AS	Assam	NORTHEAST
Tinsukia	AS	Assam	NORTHEAST

… 16 more row(s) not shown.

Theta (non-equi) join — which festival season each night of a stay falls in [joins.theta]

RA / RC: nights ⋈ night ≥ start_date ∧ night ≤ end_date seasons

The join condition is a range (BETWEEN), not equality. This is exactly how fn_quote_stay prices a stay across Diwali.

```
SELECT n.night::date, to_char(n.night, 'Dy') AS day, s.season_name, s.multiplier
FROM generate_series(DATE '2026-11-04', DATE '2026-11-12', INTERVAL '1 day') AS n(night)
JOIN seasons s ON n.night BETWEEN s.start_date AND s.end_date
ORDER BY 1
```

night	day	season_name	multiplier
2026-11-06	Fri	Diwali 2026	1.40
2026-11-07	Sat	Diwali 2026	1.40
2026-11-08	Sun	Diwali 2026	1.40
2026-11-09	Mon	Diwali 2026	1.40
2026-11-10	Tue	Diwali 2026	1.40

Left outer join — every state, even those with no hotels [joins.left_outer]

RA / RC: states ⋈ᴸ cities ⋈ᴸ hotels

Keeps every row of the left table. States with no listed hotels (Lakshadweep, Manipur…) appear with 0 — an inner join would drop them.

```
SELECT s.state_name, s.region, count(h.hotel_id) AS hotels
FROM states s
LEFT JOIN cities c ON c.state_code = s.state_code
LEFT JOIN hotels h ON h.city_id = c.city_id
GROUP BY s.state_code
ORDER BY hotels, s.state_name
```

state_name	region	hotels
Lakshadweep	SOUTH	0
Manipur	NORTHEAST	0
Arunachal Pradesh	NORTHEAST	1
Tripura	NORTHEAST	3
Dadra and Nagar Haveli and Daman and Diu	WEST	4
Mizoram	NORTHEAST	4
Nagaland	NORTHEAST	4
Meghalaya	NORTHEAST	9

… 28 more row(s) not shown.

Right outer join — every amenity, with how many hotels offer it

[joins.right_outer]

RA / RC: hotel_amenities $\bowtie^{R}$ amenities

Mirror image of the left join. Amenities nobody offers yet still appear, with 0.

```
SELECT a.category, a.amenity_name, count(ha.hotel_id) AS hotels
FROM hotel_amenities ha
RIGHT JOIN amenities a ON a.amenity_id = ha.amenity_id
GROUP BY a.amenity_id
ORDER BY hotels, a.amenity_name
```

category	amenity_name	hotels
EXPERIENCE	Beachfront	1
EXPERIENCE	Desert Safari	1
EXPERIENCE	Fireplace	1
EXPERIENCE	Guided Walks	1
EXPERIENCE	Private Deck	1
EXPERIENCE	Sea View	1
DINING	Fine Dining	2
EXPERIENCE	Lake View	3

… 44 more row(s) not shown.

Full outer join — cities where guests live vs. cities with bookings

[joins.full_outer]

RA / RC: home_cities $\bowtie^{f}$ booked_cities

Keeps unmatched rows from BOTH sides: cities that only send travellers, cities that only receive them, and those that do both. (MySQL has no FULL OUTER JOIN — one reason we chose PostgreSQL.)

```
WITH home AS (
    SELECT home_city_id AS city_id, count(*) AS guests_living_here
    FROM guests GROUP BY home_city_id
), visited AS (
    SELECT h.city_id, count(*) AS bookings_received
    FROM bookings b JOIN hotels h ON h.hotel_id = b.hotel_id
    GROUP BY h.city_id
)
SELECT c.city_name,
```

```
       coalesce(home.guests_living_here, 0) AS guests_living_here,
       coalesce(visited.bookings_received, 0) AS bookings_received,
       CASE WHEN home.city_id IS NULL THEN 'only receives'
            WHEN visited.city_id IS NULL THEN 'only sends'
            ELSE 'both' END AS role
FROM home
FULL OUTER JOIN visited ON visited.city_id = home.city_id
JOIN cities c ON c.city_id = coalesce(home.city_id, visited.city_id)
ORDER BY bookings_received DESC, guests_living_here DESC
LIMIT 20
```

city_name	guests_living_here	bookings_received	role
Delhi	882	7378	both
Bangalore	873	6891	both
Goa	171	4661	both
Chennai	864	3454	both
Mumbai	848	3277	both
Hyderabad	881	3001	both
Pune	881	2957	both
Kolkata	872	2521	both

… 12 more row(s) not shown.

Self join — who referred whom [joins.self]

RA / RC: ρ g(guests) ⋈ g.referred_by = r.guest_id ρ r(guests)

The same table appears twice under two aliases. guests.referred_by is a foreign key to guests itself.

```
SELECT g.guest_id, g.first_name || ' ' || g.last_name AS guest,
       r.guest_id AS referrer_id, r.first_name || ' ' || r.last_name AS referred_by
FROM guests g
JOIN guests r ON r.guest_id = g.referred_by
ORDER BY g.guest_id
LIMIT 15
```

guest_id	guest	referrer_id	referred_by
52	Saanvi Chouhan	38	Ayesha Thomas
56	Aanya Sen	5	Ira Parekh
65	Riya Gowda	44	Rashmi Saikia
72	Jyothi Abraham	44	Rashmi Saikia
76	Sana Gowda	61	Ishaan Parekh
86	Vignesh Yadav	66	Harjot Saha
90	Trisha Pereira	44	Rashmi Saikia
98	Ayesha Desai	47	Tanvi Rathore

… 7 more row(s) not shown.

Cross join (Cartesian product) — a price calendar [joins.cross]

RA / RC: room_types × nights

Pairs every row with every row (|R| × |S| rows). Useful when you really want all combinations: here every room type of Samudra Mahal × every night of Christmas week.

```
SELECT rt.type_name, n.night::date,
       round(rt.base_price * coalesce(s.multiplier, 1)) AS rate_before_weekend
FROM room_types rt
CROSS JOIN generate_series(DATE '2026-12-21', DATE '2026-12-27', INTERVAL '1 day') AS n(night)
LEFT JOIN seasons s ON n.night BETWEEN s.start_date AND s.end_date
WHERE rt.hotel_id = 1
ORDER BY rt.base_price, n.night
```

type_name	night	rate_before_weekend
Harbour Room	2026-12-21	27,750.00

type_name	night	rate_before_weekend
Harbour Room	2026-12-22	27,750.00
Harbour Room	2026-12-23	27,750.00
Harbour Room	2026-12-24	27,750.00
Harbour Room	2026-12-25	27,750.00
Harbour Room	2026-12-26	27,750.00
Harbour Room	2026-12-27	27,750.00
Gateway View Suite	2026-12-21	51,000.00

… 13 more row(s) not shown.

Semi-join (EXISTS) — hotels that have at least one 5★ review [joins.semi]

RA / RC: hotels ⋉ (bookings ⋈ σ rating = 5 (reviews))

Returns rows of the left table that HAVE a match, without duplicating them per match — unlike a plain join.

```
SELECT h.hotel_name, h.avg_rating, h.review_count
FROM hotels h
WHERE EXISTS (SELECT 1 FROM bookings b JOIN reviews r ON r.booking_id = b.booking_id
              WHERE b.hotel_id = h.hotel_id AND r.rating = 5)
ORDER BY h.review_count DESC
LIMIT 10
```

hotel_name	avg_rating	review_count
Samudra Mahal	4.60	155
Lutyens Lodge	4.50	148
Charminar Courtyard	4.50	136
Cubbon Leaf	4.50	118
Hooghly Ghat House	4.50	111
Spice Warehouse Fort Kochi	4.50	73
Amber Haveli	4.60	63
Pichola Rajbagh	4.40	58

… 2 more row(s) not shown.

Anti-join (NOT EXISTS) — hotels that have never been booked [joins.anti]

RA / RC: hotels ▷ bookings (= hotels − (hotels ⋉ bookings))

Rows of the left table with NO match. NOT EXISTS is NULL-safe; NOT IN (subquery) is not (one NULL makes it return nothing).

```
SELECT c.city_name, count(*) AS never_booked_hotels
FROM hotels h JOIN cities c ON c.city_id = h.city_id
WHERE NOT EXISTS (SELECT 1 FROM bookings b WHERE b.hotel_id = h.hotel_id)
GROUP BY c.city_name
ORDER BY never_booked_hotels DESC
LIMIT 10
```

city_name	never_booked_hotels
Alleppey	1
Deoghar	1

LATERAL join — the 2 cheapest room types of each signature hotel

[joins.lateral]

RA / RC: — (top-N per group; no single classical operator)

The subquery on the right can refer to the current row on the left, like a for-each loop.

```
SELECT h.hotel_name, cheapest.type_name, cheapest.base_price
FROM hotels h
CROSS JOIN LATERAL (
```

```
    SELECT rt.type_name, rt.base_price FROM room_types rt
    WHERE rt.hotel_id = h.hotel_id ORDER BY rt.base_price LIMIT 2
) AS cheapest
WHERE h.is_signature
ORDER BY h.hotel_id, cheapest.base_price
```

hotel_name	type_name	base_price
Samudra Mahal	Harbour Room	18,500.00
Samudra Mahal	Gateway View Suite	34,000.00
Charminar Courtyard	Courtyard Room	3,400.00
Charminar Courtyard	Jharokha Room	4,600.00
Pichola Rajbagh	Zenana Garden Room	24,000.00
Pichola Rajbagh	Jharokha Lake Suite	42,000.00
Morjim Driftwood Shacks	Driftwood Hut	2,900.00
Morjim Driftwood Shacks	Sea-front Hut	4,200.00

… 28 more row(s) not shown.

Six-table join — a full booking receipt [joins.multiway]

RA / RC: bookings ⋈ guests ⋈ hotels ⋈ booking_rooms ⋈ rooms ⋈ room_types

Follows the foreign keys from a booking to everything it touches. This is what the 'My trips' page needs.

```
SELECT b.booking_ref, g.first_name || ' ' || g.last_name AS guest, h.hotel_name,
       r.room_number, rt.type_name, b.check_in, b.check_out, b.nights,
       br.room_subtotal, br.room_tax, b.status
FROM bookings b
JOIN guests g        ON g.guest_id = b.guest_id
JOIN hotels h        ON h.hotel_id = b.hotel_id
JOIN booking_rooms br ON br.booking_id = b.booking_id
JOIN rooms r         ON r.room_id = br.room_id
JOIN room_types rt   ON rt.room_type_id = r.room_type_id
WHERE b.hotel_id = 1 AND b.check_in >= DATE '2026-12-01'
ORDER BY b.check_in, r.room_number
LIMIT 15
```

booking_ref	guest	hotel_name	room_number	type_name	check_in	check_out	nights	room_subtotal	room_tax	status
YRFYBF2M	Jacob Trivedi	Samudra Mahal	110	Harbour Room	2026-12-01	2026-12-02	1	18,500.00	3,330.00	CONFIRMED
YRQQWSF8	Aishwarya Rao	Samudra Mahal	204	Gateway View Suite	2026-12-03	2026-12-13	10	353,600.00	63,648.00	CONFIRMED
YR8DAFPA	Saurabh Subramanian	Samudra Mahal	203	Gateway View Suite	2026-12-04	2026-12-09	5	176,800.00	31,824.00	CONFIRMED
YRTBGYQH	Swati Banerjee	Samudra Mahal	205	Gateway View Suite	2026-12-05	2026-12-07	2	71,400.00	12,852.00	CONFIRMED
YRTBGYQH	Swati Banerjee	Samudra Mahal	208	Gateway View Suite	2026-12-05	2026-12-07	2	71,400.00	12,852.00	CONFIRMED
YRGAWFWR	Sneha Sangma	Samudra Mahal	209	Bunder Presidential Suite	2026-12-05	2026-12-07	2	151,200.00	27,216.00	CONFIRMED
YR66VAXC	Gurpreet Verma	Samudra Mahal	210	Bunder Presidential Suite	2026-12-05	2026-12-08	3	223,200.00	40,176.00	CONFIRMED
YR6JQBHF	Gaurav Kulkarni	Samudra Mahal	108	Harbour Room	2026-12-09	2026-12-11	2	37,000.00	6,660.00	CONFIRMED

… 7 more row(s) not shown.

UNION — every place name a guest can type into search [sets.union]

RA / RC: π name (cities) ∪ π name (landmarks)

Both inputs must be union-compatible (same number and types of columns). UNION removes duplicates; UNION ALL keeps them and is faster.

```
SELECT city_name AS place, 'city' AS kind FROM cities WHERE city_name ILIKE 'ma%'
UNION
SELECT landmark_name, 'landmark' FROM landmarks WHERE landmark_name ILIKE 'ma%'
ORDER BY place
```

place	kind
Madan Mahal Fort	landmark
Madan Mohan temple	landmark
Madanmahal Station	landmark
Madgaon Railway Station	landmark
Madgoan Railway Station	landmark
Madhatop	landmark
Madikeri	landmark
Madikeri Fort	landmark

... 114 more row(s) not shown.

INTERSECT — guests who stayed in both Goa and Mumbai [sets.intersect]

RA / RC: π guest_id (σ city='Goa' (...)) ∩ π guest_id (σ city='Mumbai' (...))

Rows present in both results. (MySQL only added INTERSECT in 8.0.31.)

```
SELECT b.guest_id FROM bookings b JOIN hotels h USING (hotel_id) JOIN cities c USING (city_id)
WHERE c.city_name = 'Goa' AND b.status = 'COMPLETED'
INTERSECT
SELECT b.guest_id FROM bookings b JOIN hotels h USING (hotel_id) JOIN cities c USING (city_id)
WHERE c.city_name = 'Mumbai' AND b.status = 'COMPLETED'
ORDER BY guest_id
LIMIT 20
```

guest_id
1
2
3
4
5
6
8
9

... 12 more row(s) not shown.

EXCEPT (set difference) — signature hotels with no pool [sets.except]

RA / RC: π hotel_id (σ is_signature (hotels)) − π hotel_id (hotel_amenities ⋈ σ name='Swimming Pool' (amenities))

Rows in the first result that are not in the second.

```
SELECT h.hotel_id, h.hotel_name FROM hotels h WHERE h.is_signature
EXCEPT
SELECT ha.hotel_id, h.hotel_name FROM hotel_amenities ha
JOIN amenities a USING (amenity_id) JOIN hotels h USING (hotel_id)
WHERE a.amenity_name = 'Swimming Pool'
ORDER BY hotel_id
```

hotel_id	hotel_name
2	Charminar Courtyard
4	Morjim Driftwood Shacks
6	Vembanad Kettuvallam
7	Ghat Ninety-One
8	Chai Bagaan Bungalow
9	Dal Shikara House
10	Sonar Qila Haveli
11	Lutyens Lodge

… 5 more row(s) not shown.

GROUP BY + HAVING — cities with more than 50 hotels [agg.group_having]

RA / RC: σ hotels > 50 (city_name γ count(hotel_id) → hotels, avg(star_rating) (hotels ⋈ cities))

WHERE filters rows before grouping; HAVING filters groups after.

```
SELECT c.city_name, count(*) AS hotels, round(avg(h.star_rating), 2) AS avg_stars,
       min(v.from_price) AS cheapest_from
FROM hotels h JOIN cities c USING (city_id) JOIN v_hotel_cards v USING (hotel_id)
GROUP BY c.city_name
HAVING count(*) > 50
ORDER BY hotels DESC
```

city_name	hotels	avg_stars	cheapest_from
Goa	200	0.89	499.00
Bangalore	131	1.10	499.00
Delhi	120	1.78	499.00
Gurgaon	80	1.00	899.00
Alleppey	69	0.70	899.00
Manali	68	1.47	899.00
Chennai	67	1.31	499.00
Coorg	65	0.60	599.00

… 7 more row(s) not shown.

ROLLUP — revenue by region → state, with subtotals and grand total

[agg.rollup]

RA / RC: — (OLAP extension of γ)

ROLLUP(a, b) = GROUP BY (a,b) ∪ (a) ∪ (). NULL in the output marks a subtotal row.

```
SELECT coalesce(s.region::text, '— ALL REGIONS —') AS region,
       coalesce(s.state_name, '— subtotal —')      AS state,
       count(DISTINCT b.booking_id)                AS bookings,
       sum(br.room_subtotal)                       AS room_revenue
FROM bookings b
JOIN booking_rooms br USING (booking_id)
JOIN hotels h USING (hotel_id) JOIN cities c USING (city_id) JOIN states s USING (state_code)
WHERE b.status = 'COMPLETED'
GROUP BY ROLLUP (s.region, s.state_name)
ORDER BY s.region NULLS LAST, room_revenue DESC NULLS FIRST
```

region	state	bookings	room_revenue
NORTH	— subtotal —	13462	221,569,841.00
NORTH	Delhi	5112	105,693,721.00
NORTH	Uttar Pradesh	2136	24,761,809.00
NORTH	Uttarakhand	1470	24,305,442.00
NORTH	Haryana	1588	20,029,871.00

region	state	bookings	room_revenue
NORTH	Himachal Pradesh	1271	17,852,163.00
NORTH	Jammu and Kashmir	696	12,395,672.00
NORTH	Chandigarh	576	9,443,579.00

... 33 more row(s) not shown.

CUBE — bookings by status × payment method [agg.cube]

RA / RC: — (OLAP extension of γ)

CUBE(a, b) produces every combination of subtotals: (a,b), (a), (b) and ().

```
SELECT coalesce(b.status::text, 'ALL') AS status,
       coalesce(p.method::text, 'ALL') AS method,
       count(*) AS payments
FROM payments p JOIN bookings b USING (booking_id)
WHERE p.kind = 'CHARGE' AND p.status = 'SUCCESS'
GROUP BY CUBE (b.status, p.method)
ORDER BY b.status NULLS LAST, p.method NULLS LAST
```

status	method	payments
CONFIRMED	UPI	6471
CONFIRMED	CARD	2992
CONFIRMED	NETBANKING	1140
CONFIRMED	WALLET	588
CONFIRMED	ALL	11191
CHECKED_IN	UPI	168
CHECKED_IN	CARD	96
CHECKED_IN	NETBANKING	32

... 25 more row(s) not shown.

Division (÷) — hotels that have ALL of Pool, Spa and Free Wi-Fi (double NOT EXISTS) [division.not_exists]

RA / RC: π hotel_id, amenity_id (hotel_amenities) ÷ π amenity_id (σ name ∈ {Pool, Spa, Wi-Fi} (amenities))

"Hotels for which there is NO required amenity that they do NOT have." This is the SQL form of the TRC formula {h | ∀a (a ∈ Req → ∃ha (ha.hotel=h ∧ ha.amenity=a))}.

```
SELECT h.hotel_id, h.hotel_name, c.city_name
FROM hotels h JOIN cities c USING (city_id)
WHERE NOT EXISTS (
    SELECT 1 FROM amenities a
    WHERE a.amenity_name IN ('Swimming Pool', 'Spa', 'Free Wi-Fi')
      AND NOT EXISTS (SELECT 1 FROM hotel_amenities ha
                      WHERE ha.hotel_id = h.hotel_id AND ha.amenity_id = a.amenity_id))
ORDER BY c.city_name, h.hotel_name
```

hotel_id	hotel_name	city_name
2047	Hotel Pushp Villa	Agra
2199	The Retreat	Agra
202	BigSplash	Alibag
2655	The World Backwaters	Alleppey
1335	Boulevard Riviera	Bangalore
554	DEEYASUITES	Bangalore
2487	The Waverly Hotel & Residences	Bangalore
2460	Urban Stay	Bangalore

... 88 more row(s) not shown.

Division (÷) — the same answer with GROUP BY / HAVING COUNT

[division.count]

RA / RC: same as division.not_exists

Keep each hotel's rows for the required amenities and accept it only if it matched all 3. Usually faster than double NOT EXISTS.

```
SELECT h.hotel_id, h.hotel_name, c.city_name
FROM hotel_amenities ha
JOIN amenities a USING (amenity_id)
JOIN hotels h USING (hotel_id) JOIN cities c USING (city_id)
WHERE a.amenity_name IN ('Swimming Pool', 'Spa', 'Free Wi-Fi')
GROUP BY h.hotel_id, h.hotel_name, c.city_name
HAVING count(DISTINCT a.amenity_id) = 3
ORDER BY c.city_name, h.hotel_name
```

hotel_id	hotel_name	city_name
2047	Hotel Pushp Villa	Agra
2199	The Retreat	Agra
202	BigSplash	Alibag
2655	The World Backwaters	Alleppey
1335	Boulevard Riviera	Bangalore
554	DEEYASUITES	Bangalore
2487	The Waverly Hotel & Residences	Bangalore
2460	Urban Stay	Bangalore

… 88 more row(s) not shown.

Scalar subquery — room types priced above their city's average

[sub.scalar]

RA / RC: σ base_price > avg_city (room_types ⋈ hotels)

The inner query is re-evaluated for each outer row (correlated) because it refers to h.city_id.

```
SELECT h.hotel_name, c.city_name, rt.type_name, rt.base_price,
       (SELECT round(avg(rt2.base_price)) FROM room_types rt2 JOIN hotels h2 USING (hotel_id)
        WHERE h2.city_id = h.city_id) AS city_avg
FROM room_types rt JOIN hotels h USING (hotel_id) JOIN cities c USING (city_id)
WHERE c.city_name = 'Udaipur'
  AND rt.base_price > (SELECT avg(rt2.base_price) FROM room_types rt2 JOIN hotels h2 USING (hotel_id)
                       WHERE h2.city_id = h.city_id)
ORDER BY rt.base_price DESC
LIMIT 10
```

hotel_name	city_name	type_name	base_price	city_avg
Pichola Rajbagh	Udaipur	Maharani Suite	95,000.00	7,560.00
Pichola Rajbagh	Udaipur	Jharokha Lake Suite	42,000.00	7,560.00
THe Udai bagh	Udaipur	Junior Suite	28,299.00	7,560.00
Hotel Raj Niwas	Udaipur	Super Deluxe Room	25,499.00	7,560.00
The Royal Retreat Resort and Spa	Udaipur	Junior Suite	24,799.00	7,560.00
Pichola Rajbagh	Udaipur	Zenana Garden Room	24,000.00	7,560.00
THe Udai bagh	Udaipur	Premium Room	20,599.00	7,560.00
Hotel Raj Niwas	Udaipur	Deluxe Room	19,699.00	7,560.00

… 2 more row(s) not shown.

> ALL — rooms dearer than every room in Varanasi [sub.any_all]

RA / RC: σ price > max(π price (σ city='Varanasi' …))

x > ALL (S) is true when x beats every element of S; x > ANY (S) when it beats at least one.

```
SELECT h.hotel_name, rt.type_name, rt.base_price
FROM room_types rt JOIN hotels h USING (hotel_id)
WHERE rt.base_price > ALL (SELECT rt2.base_price FROM room_types rt2
                           JOIN hotels h2 USING (hotel_id) JOIN cities c2 USING (city_id)
                           WHERE c2.city_name = 'Varanasi')
ORDER BY rt.base_price
LIMIT 10
```

hotel_name	type_name	base_price
Ganga Kinara Retreat	River Cottage	7,900.00
Signature Crest Cochin	Premium Room	7,999.00
Breeze Residency	Super Deluxe Room	7,999.00
White Ridge Hotel	Deluxe Room	7,999.00
Blueberry Inn	Super Deluxe Room	7,999.00
Elysium Resorts	Deluxe Room	7,999.00
Chumbi Residency	Deluxe Room	7,999.00
The Willow Resorts	Deluxe Room	7,999.00

… 2 more row(s) not shown.

CTE (WITH) — each city's busiest hotel [sub.cte]

RA / RC: — (named sub-expressions)

A CTE names an intermediate result so a long query reads top-to-bottom.

```
WITH per_hotel AS (
    SELECT h.city_id, h.hotel_id, h.hotel_name, count(*) AS bookings
    FROM bookings b JOIN hotels h USING (hotel_id)
    WHERE b.status = 'COMPLETED'
    GROUP BY h.hotel_id
), ranked AS (
    SELECT *, rank() OVER (PARTITION BY city_id ORDER BY bookings DESC) AS rnk FROM per_hotel
)
SELECT c.city_name, r.hotel_name, r.bookings
FROM ranked r JOIN cities c USING (city_id)
WHERE r.rnk = 1
ORDER BY r.bookings DESC
LIMIT 15
```

city_name	hotel_name	bookings
Mumbai	Samudra Mahal	396
Delhi	Lutyens Lodge	369
Hyderabad	Charminar Courtyard	315
Bangalore	Cubbon Leaf	302
Kolkata	Hooghly Ghat House	276
Kochi	Spice Warehouse Fort Kochi	190
Jaipur	Amber Haveli	187
Udaipur	Pichola Rajbagh	179

… 7 more row(s) not shown.

Recursive CTE — the full referral chain above a guest [sub.recursive]

RA / RC: — (transitive closure is NOT expressible in basic relational algebra)

Starts from one guest and repeatedly joins to the referrer until there is none. Classic recursive relationship.

```
WITH RECURSIVE chain AS (
    SELECT guest_id, first_name, last_name, referred_by, 0 AS depth
    FROM guests
    WHERE guest_id = (SELECT g.guest_id FROM guests g
                      JOIN guests p ON p.guest_id = g.referred_by
                      JOIN guests gp ON gp.guest_id = p.referred_by
                      WHERE gp.referred_by IS NOT NULL
                      ORDER BY g.guest_id LIMIT 1)
```

```
  UNION ALL
    SELECT g.guest_id, g.first_name, g.last_name, g.referred_by, c.depth + 1
    FROM guests g JOIN chain c ON g.guest_id = c.referred_by
)
SELECT depth, guest_id, first_name || ' ' || last_name AS guest, referred_by
FROM chain ORDER BY depth
```

depth	guest_id	guest	referred_by
0	525	Suresh Mehra	303
1	303	Debashish Mukherjee	72
2	72	Jyothi Abraham	44
3	44	Rashmi Saikia	NULL

Tuple relational calculus — guests who reviewed a Udaipur stay with 5★

[calculus.trc]

RA / RC: { g.first_name | guests(g) ∧ ∃b ∃r ∃h ∃c (bookings(b) ∧ reviews(r) ∧ hotels(h) ∧ cities(c) ∧ b.guest_id = g.guest_id ∧ r.booking_id = b.booking_id ∧ r.rating = 5 ∧ b.hotel_id = h.hotel_id ∧ h.city_id = c.city_id ∧ c.city_name = 'Udaipur') }

TRC variables range over TUPLES. "∃" becomes EXISTS, "∧" becomes AND. SQL itself is based on TRC.

```
SELECT g.guest_id, g.first_name, g.last_name
FROM guests g
WHERE EXISTS (SELECT 1 FROM bookings b, reviews r, hotels h, cities c
              WHERE b.guest_id = g.guest_id AND r.booking_id = b.booking_id AND r.rating = 5
                AND b.hotel_id = h.hotel_id AND h.city_id = c.city_id AND c.city_name = 'Udaipur')
ORDER BY g.guest_id
LIMIT 15
```

guest_id	first_name	last_name
26	Pooja	Iyengar
27	Anirban	Jha
30	Manish	Abraham
41	Akash	Malhotra
47	Tanvi	Rathore
52	Saanvi	Chouhan
54	Jyothi	Shetty
65	Riya	Gowda

… 7 more row(s) not shown.

Domain relational calculus — names and stars of Goa beach huts

[calculus.drc]

RA / RC: { ⟨n, s⟩ | ∃i ∃t ∃c (hotels(i, n, t, s, c) ∧ t = 'Beach Hut' ∧ ∃cn (cities(c, cn) ∧ cn = 'Goa')) }

DRC variables range over attribute VALUES (domains), not whole tuples. Each variable here is a column.

```
SELECT n AS hotel_name, s AS star_rating
FROM (SELECT hotel_id AS i, hotel_name AS n, property_type AS t, star_rating AS s, city_id AS c FROM hotels) AS
    hotels_dom
WHERE t IN ('Beach Hut', 'Cottage')
  AND EXISTS (SELECT 1 FROM (SELECT city_id AS c2, city_name AS cn FROM cities) cities_dom
              WHERE cities_dom.c2 = hotels_dom.c AND cn = 'Goa')
ORDER BY n
```

hotel_name	star_rating
Bhau Cottages	0
Cabo Cabana Wooden Cottages	0
Casa D Calma	0
Cuba Agonda	2

hotel_name	star_rating
Cuba Beach Huts	0
Morjim Driftwood Shacks	0
Sunway Cottagesb Goa	0
Tivai Beach Cottages	0

… 3 more row(s) not shown.

Window functions — monthly revenue with running total and month-on-month change [sub.window]

RA / RC: — (OLAP)

Window functions compute across related rows without collapsing them, unlike GROUP BY.

```
WITH monthly AS (
    SELECT date_trunc('month', b.check_in)::date AS month, sum(br.room_subtotal) AS revenue
    FROM bookings b JOIN booking_rooms br USING (booking_id)
    WHERE b.status IN ('COMPLETED', 'CHECKED_IN')
    GROUP BY 1
)
SELECT month, revenue,
       sum(revenue) OVER (ORDER BY month)                       AS running_total,
       round(100.0 * (revenue / lag(revenue) OVER (ORDER BY month) - 1), 1) AS mom_change_pct,
       rank() OVER (ORDER BY revenue DESC)                      AS revenue_rank
FROM monthly ORDER BY month
```

month	revenue	running_total	mom_change_pct	revenue_rank
2025-01-01	19,145,395.00	19,145,395.00	NULL	19
2025-02-01	14,400,834.00	33,546,229.00	-24.80	21
2025-03-01	22,272,919.00	55,819,148.00	54.70	17
2025-04-01	19,903,048.00	75,722,196.00	-10.60	18
2025-05-01	36,796,085.00	112,518,281.00	84.90	10
2025-06-01	33,425,310.00	145,943,591.00	-9.20	13
2025-07-01	12,905,018.00	158,848,609.00	-61.40	22
2025-08-01	14,710,060.00	173,558,669.00	14.00	20

… 14 more row(s) not shown.

9. Retrieval: what the website asks the database

Price for two nights — Samudra Mahal Harbour Room, 24 → 26 Dec 2026

[app.quote_two_nights]

RA / RC: room_types × nights ⋈ᴸ seasons (inside fn_quote_stay)

Night-by-night breakdown: base price × festival multiplier × weekend multiplier, then GST slab per night.

```
SELECT * FROM fn_quote_stay(1, DATE '2026-12-24', DATE '2026-12-26')
```

night	day_name	season_name	base_price	season_multiplier	weekend_multiplier	nightly_rate	gst_rate	gst_amount	rooms	night_total
2026-12-24	Thu	Christmas & New Year 2026	18,500.00	1.50	1.00	27,750.00	0.18	4,995.00	1	32,745.00
2026-12-25	Fri	Christmas & New Year 2026	18,500.00	1.50	1.10	30,525.00	0.18	5,494.50	1	36,019.50

Same room, three different weekends — why prices differ

[app.quote_compare_dates]

RA / RC: dates ⋈ fn_stay_total(…)

The same 2-night stay costs more in Diwali and Christmas week (season multiplier) than in the monsoon.

```
SELECT d.label, d.check_in, d.check_in + 2 AS check_out,
       fn_stay_total(1, d.check_in, d.check_in + 2) AS total_incl_gst
FROM (VALUES ('Monsoon weekend', DATE '2026-08-14'),
             ('Ordinary weekend', DATE '2026-11-20'),
             ('Diwali weekend',   DATE '2026-11-06'),
             ('Christmas',        DATE '2026-12-25')) AS d(label, check_in)
ORDER BY total_incl_gst
```

label	check_in	check_out	total_incl_gst
Monsoon weekend	2026-08-14	2026-08-16	40,823.28
Ordinary weekend	2026-11-20	2026-11-22	48,026.00
Diwali weekend	2026-11-06	2026-11-08	67,236.40
Christmas	2026-12-25	2026-12-27	72,039.00

Availability — free rooms per type at Pichola Rajbagh for Diwali

[app.availability]

RA / RC: rooms ▷ (booking_rooms ⋈ σ overlaps(stay) ∧ status ≠ CANCELLED (bookings))

A room is free if no live booking overlaps the requested half-open range [check_in, check_out).

```
SELECT * FROM fn_available_rooms(3, DATE '2026-11-07', DATE '2026-11-09')
```

room_type_id	type_name	base_price	max_adults	max_children	total_rooms	free_rooms
7	Zenana Garden Room	24,000.00	2	1	10	7
8	Jharokha Lake Suite	42,000.00	3	1	5	5
9	Maharani Suite	95,000.00	4	2	1	1

"Near Gateway of India" — hotels within 3 km, nearest first

[app.near_landmark]

RA / RC: π … (σ landmark_name ILIKE … ∧ distance_km ≤ 3 (landmarks ⋈ hotel_landmarks ⋈ hotels))

The M:N table hotel_landmarks carries the distance, an attribute of the relationship itself.

```
SELECT h.hotel_name, h.star_rating, hl.distance_km, v.from_price
FROM landmarks l
JOIN hotel_landmarks hl USING (landmark_id)
JOIN hotels h USING (hotel_id)
JOIN v_hotel_cards v USING (hotel_id)
WHERE l.landmark_name ILIKE '%gateway of india%' AND hl.distance_km <= 3
ORDER BY hl.distance_km
LIMIT 15
```

hotel_name	star_rating	distance_km	from_price
Samudra Mahal	5	0.23	18,500.00
Astoria Hotel	3	1.30	6,099.00
Traveller's Inn Hotel	2	1.60	3,799.00
West End Hotel	4	2.20	10,299.00
Hotel Sea Lord	2	2.81	3,699.00
Dreamland	1	2.95	3,199.00

Mood search — "quiet hill stay" with full-text ranking [app.mood_search]

RA / RC: — (ranked retrieval)

to_tsquery matches word stems (hills → hill) through the GIN index; ts_rank orders by relevance like a search engine.

```
SELECT h.hotel_name, c.city_name, h.property_type,
       round(ts_rank(h.search_doc, q)::numeric, 3) AS relevance
FROM hotels h JOIN cities c USING (city_id),
     websearch_to_tsquery('english', 'quiet hill stay') AS q
WHERE h.search_doc @@ q
ORDER BY relevance DESC, h.avg_rating DESC NULLS LAST
LIMIT 12
```

hotel_name	city_name	property_type	relevance
Pine Needle Cottage	Shimla	Cottage	0.79
Chilipili Estate Stay	Coorg	Homestay	0.66
Daiwik Home Stay	Coorg	Homestay	0.66
Deepa home stay	Coorg	Homestay	0.66
Vardaan Home Stay	Shimla	Homestay	0.66
Sweet Home Stay	Coorg	Homestay	0.66
yamini home stay	Kasauli	Homestay	0.66
Tulip Estate Stay	Coorg	Homestay	0.66
Chingaara Estate Stay	Coorg	Homestay	0.66
Coorg home stay	Coorg	Homestay	0.66

… 2 more row(s) not shown.

Typo-tolerant search — "samudra mahl" [app.fuzzy_name]

RA / RC: σ similarity(hotel_name, 'samudra mahl') > 0.3 (hotels)

pg_trgm splits strings into 3-letter pieces; the GIN trigram index finds close matches even with typos.

```
SELECT hotel_name, round(similarity(hotel_name, 'samudra mahl')::numeric, 2) AS similarity
FROM hotels
WHERE hotel_name % 'samudra mahl'
ORDER BY similarity DESC
LIMIT 5
```

hotel_name	similarity
Samudra Mahal	0.69
HOTEL SAMUDRA	0.42

hotel_name	similarity
Hotel Samudra City	0.33

Occupancy of signature hotels on Diwali night (8 Nov 2026) [app.occupancy]

RA / RC: hotels γ count(rooms booked) / count(rooms)

Rooms in use on a night = live bookings whose range contains that night.

```
SELECT h.hotel_name,
       count(DISTINCT r.room_id) AS rooms,
       count(DISTINCT br.room_id) AS occupied,
       round(100.0 * count(DISTINCT br.room_id) / count(DISTINCT r.room_id), 1) AS occupancy_pct
FROM hotels h
JOIN rooms r ON r.hotel_id = h.hotel_id
LEFT JOIN booking_rooms br ON br.room_id = r.room_id
     AND EXISTS (SELECT 1 FROM bookings b WHERE b.booking_id = br.booking_id
                 AND b.status <> 'CANCELLED'
                 AND DATE '2026-11-08' >= b.check_in AND DATE '2026-11-08' < b.check_out)
WHERE h.is_signature
GROUP BY h.hotel_id
ORDER BY occupancy_pct DESC
```

hotel_name	rooms	occupied	occupancy_pct
Lutyens Lodge	18	16	88.90
Hooghly Ghat House	7	6	85.70
Pine Needle Cottage	3	2	66.70
Ghat Ninety-One	11	5	45.50
Rue Romain Villa	7	3	42.90
Amber Haveli	15	6	40.00
Vembanad Kettuvallam	5	2	40.00
Charminar Courtyard	11	4	36.40
Samudra Mahal	22	7	31.80
Spice Warehouse Fort Kochi	14	2	14.30

… 8 more row(s) not shown.

Top 10 cities by room revenue (materialized view) [app.top_cities]

RA / RC: π … (mv_city_performance)

Reads precomputed results — instant, but only as fresh as the last REFRESH MATERIALIZED VIEW.

```
SELECT city_name, state_code, hotels, bookings, room_nights, room_revenue, avg_daily_rate
FROM mv_city_performance
ORDER BY room_revenue DESC
LIMIT 10
```

city_name	state_code	hotels	bookings	room_nights	room_revenue	avg_daily_rate
Delhi	DL	120	5160	17,026.00	106,868,795.00	6,277.00
Mumbai	MH	52	2288	7,439.00	83,566,533.00	11,234.00
Bangalore	KA	131	4858	16,138.00	72,183,855.00	4,473.00
Goa	GA	200	3234	10,566.00	56,950,761.00	5,390.00
Chennai	TN	67	2375	7,870.00	40,290,805.00	5,120.00
Udaipur	RJ	41	989	3,284.00	38,896,547.00	11,844.00
Pune	MH	57	2082	6,844.00	36,187,856.00	5,288.00
Jaipur	RJ	63	1386	4,608.00	33,691,446.00	7,312.00
Kolkata	WB	41	1760	5,861.00	31,872,169.00	5,438.00
Ahmedabad	GJ	51	1760	6,002.00	30,912,868.00	5,150.00

9.1 'Why can't I…?' — every refusal is a named rule

attempt	mechanism	database message	rule
Why can't I check out before I check in?	CHECK constraint (tuple-level)	new row for relation "bookings" violates check constraint "ck_bookings_dates"	CHECK constraint: check-out must be after check-in.
Why can't I book 45 nights?	CHECK constraint (business limit)	new row for relation "bookings" violates check constraint "ck_bookings_max_stay"	CHECK constraint: a booking can be at most 30 nights.
Why can't two guests get the same room on the same night?	Trigger (multi-row rule)	Room 110 is already booked (YRKPS8FJ) for an overlapping stay	TRIGGER trg_booking_rooms_guard: a room must belong to the booking's hotel and be free for those dates.
Why can't 9 adults share one Harbour Room?	Deferred constraint trigger (checked at COMMIT)	Booking YRUYDD5K has 9 adults + 0 children but the rooms hold 2 adults + 1 children	DEFERRED CONSTRAINT TRIGGER trg_booking_capacity: too many guests for the rooms chosen.
Why can't I book 3 Maharani Suites?	Stored function with row locks (SELECT … FOR UPDATE)	Only 1 room(s) of this type are free from 2026-12-28 to 2026-12-30	FUNCTION fn_book_stay: not enough free rooms (rows were locked with SELECT … FOR UPDATE).
Why can't a completed stay be moved back to CONFIRMED?	Trigger (state machine)	A COMPLETED booking cannot become CONFIRMED	TRIGGER trg_bookings_status: that status change is not allowed.
Why can't I delete a hotel that has bookings?	Foreign key · ON DELETE RESTRICT	update or delete on table "hotels" violates foreign key constraint "fk_bookings_hotels" on table "bookings"	FOREIGN KEY (ON DELETE RESTRICT): the hotel still has bookings.
Why is 12345 67890 not accepted as a phone number?	DOMAIN constraint	value for domain indian_mobile violates check constraint "indian_mobile_check"	DOMAIN indian_mobile: 10 digits, starting with 6, 7, 8 or 9.
Why can't I sign up with a phone that is already registered?	UNIQUE constraint (candidate key)	duplicate key value violates unique constraint "uq_guests_phone"	UNIQUE (candidate key): this phone number already belongs to another guest.
Why can't a 15-year-old create an account?	CHECK constraint using a date expression	new row for relation "guests" violates check constraint "ck_guests_adult"	CHECK constraint: a guest must be at least 18 years old.
Why can't I review a stay that hasn't happened yet?	Trigger (cross-table rule)	Booking YRW474CC is CONFIRMED; only completed stays can be reviewed	TRIGGER trg_reviews_guard: only completed stays can be reviewed, after check-out.
Why can't I review the same stay twice?	UNIQUE on a foreign key (1:1 relationship)	duplicate key value violates unique constraint "uq_reviews_booking"	UNIQUE: a stay can be reviewed only once (1:1 bookings ↔ reviews).
Why can't I add a "Wedding Season" that overlaps Diwali?	EXCLUDE constraint	conflicting key value violates exclusion constraint "ex_seasons_no_overlap"	EXCLUDE constraint: two pricing seasons cannot overlap.
Why can't I refund more than the guest paid?	Trigger (aggregate rule)	Refund ₹7385.70 is more than the ₹3692.85 paid for this booking	TRIGGER trg_payments_refund_guard: refund larger than amount paid.
Why can't a Mumbai booking use a room in Udaipur?	Trigger + composite key consistency	Room 101 belongs to hotel 3, but booking YRJEWF2X is at hotel 1	TRIGGER trg_booking_rooms_guard: a room must belong to the booking's hotel and be free for those dates.
Why can't the analyst read guests' phone numbers?	DCL · GRANT / REVOKE · security views	permission denied for table guests	PRIVILEGES (GRANT / REVOKE): this database role is not allowed to do that.

10. Query processing and optimization

PostgreSQL translates SQL into a tree of relational-algebra operators and chooses an algorithm for each using cost estimates from table statistics (ANALYZE). The plans below were produced by EXPLAIN (ANALYZE, BUFFERS) on the project data; '@setup' settings disable alternatives so each algorithm can be seen.

SELECT with no usable index → sequential (linear) scan

Algorithms for SELECT › Linear search

special_request has no index, so every block of bookings is read and each row is tested.

```
SELECT booking_ref FROM bookings WHERE special_request = 'Celebrating anniversary'
```

```
Seq Scan on bookings  (cost=0.00..1796.00 rows=557 width=9) (actual time=0.005..3.421 rows=556 loops=1)
  Filter: ((special_request)::text = 'Celebrating anniversary'::text)
  Rows Removed by Filter: 71446
  Buffers: shared hit=896
Planning:
  Buffers: shared hit=3
Planning Time: 0.017 ms
Execution Time: 3.434 ms
```

operators: Seq Scan · execution 3.515 ms

SELECT on a key → B-tree index scan

Algorithms for SELECT › Index search

booking_ref has a UNIQUE index: the B-tree is descended in a few page reads instead of scanning 72,000 rows.

```
SELECT * FROM bookings WHERE booking_ref = (SELECT booking_ref FROM bookings WHERE booking_id = 500)
```

```
Index Scan using uq_bookings_ref on bookings  (cost=8.73..16.75 rows=1 width=80) (actual time=0.011..0.011 rows=
    1 loops=1)
  Index Cond: (booking_ref = $0)
  Buffers: shared hit=7
  InitPlan 1 (returns $0)
    ->  Index Scan using pk_bookings on bookings bookings_1  (cost=0.29..8.31 rows=1 width=9) (actual time=0.001
    ..0.002 rows=1 loops=1)
          Index Cond: (booking_id = 500)
          Buffers: shared hit=3
Planning:
  Buffers: shared hit=12
Planning Time: 0.031 ms
Execution Time: 0.015 ms
```

operators: Index Scan · execution 0.006 ms

SELECT on a range → bitmap index scan

Algorithms for SELECT › Index for range / many rows

Many matching rows: the index builds a bitmap of pages first, then each page is read once in physical order.

```
SELECT booking_ref, status FROM bookings WHERE hotel_id = 1 AND check_in BETWEEN DATE '2026-01-01' AND DATE '202
    6-06-30'
```

```
Bitmap Heap Scan on bookings  (cost=6.61..459.45 rows=182 width=13) (actual time=0.018..0.049 rows=177 loops=1)
  Recheck Cond: ((hotel_id = 1) AND (check_in >= '2026-01-01'::date) AND (check_in <= '2026-06-30'::date))
  Heap Blocks: exact=128
  Buffers: shared hit=131
  ->  Bitmap Index Scan on ix_bookings_hotel_dates  (cost=0.00..6.57 rows=182 width=0) (actual time=0.009..0.009
     rows=177 loops=1)
        Index Cond: ((hotel_id = 1) AND (check_in >= '2026-01-01'::date) AND (check_in <= '2026-06-30'::date))
        Buffers: shared hit=3
Planning Time: 0.018 ms
Execution Time: 0.056 ms
```

operators: Bitmap Heap Scan, Bitmap Index Scan · execution 0.043 ms

JOIN → hash join (the optimizer's default for big equi-joins)

Algorithms for JOIN › Hash join

Build a hash table on the smaller input (hotels), then probe it once per booking row. O(|R| + |S|).

```
SELECT h.property_type, count(*) FROM bookings b JOIN hotels h ON h.hotel_id = b.hotel_id GROUP BY 1
```

```
HashAggregate  (cost=4094.84..4095.05 rows=21 width=12) (actual time=13.322..13.324 rows=21 loops=1)
  Group Key: h.property_type
  Batches: 1  Memory Usage: 24kB
  Buffers: shared hit=2745
  ->  Hash Join  (cost=1929.62..3734.84 rows=72000 width=4) (actual time=1.068..9.164 rows=72002 loops=1)
        Hash Cond: (b.hotel_id = h.hotel_id)
        Buffers: shared hit=2745
        ->  Seq Scan on bookings b  (cost=0.00..1616.00 rows=72000 width=4) (actual time=0.001..2.048 rows=72002
     loops=1)
              Buffers: shared hit=896
        ->  Hash  (cost=1884.83..1884.83 rows=3583 width=8) (actual time=1.065..1.065 rows=3583 loops=1)
              Buckets: 4096  Batches: 1  Memory Usage: 172kB
              Buffers: shared hit=1849
              ->  Seq Scan on hotels h  (cost=0.00..1884.83 rows=3583 width=8) (actual time=0.001..0.901 rows=35
    83 loops=1)
                    Buffers: shared hit=1849
Planning:
  Buffers: shared hit=13
Planning Time: 0.066 ms
Execution Time: 13.335 ms
```

operators: Aggregate, Hash, Hash Join (Inner), Seq Scan · execution 12.816 ms

Same JOIN, hash join disabled → sort-merge join

Algorithms for JOIN › Sort-merge join

Both inputs are sorted on hotel_id (or read in index order) and merged in one pass, like merging two sorted lists.
setup: SET LOCAL enable_hashjoin = off; SET LOCAL enable_nestloop = off

```
SELECT h.property_type, count(*) FROM bookings b JOIN hotels h ON h.hotel_id = b.hotel_id GROUP BY 1
```

```
Finalize GroupAggregate  (cost=5940.25..5945.57 rows=21 width=12) (actual time=7.800..8.537 rows=21 loops=1)
  Group Key: h.property_type
  Buffers: shared hit=5932 dirtied=1
  ->  Gather Merge  (cost=5940.25..5945.15 rows=42 width=12) (actual time=7.797..8.530 rows=52 loops=1)
        Workers Planned: 2
        Workers Launched: 2
        Buffers: shared hit=5932 dirtied=1
        ->  Sort  (cost=4940.23..4940.28 rows=21 width=12) (actual time=5.522..5.523 rows=17 loops=3)
              Sort Key: h.property_type
              Sort Method: quicksort  Memory: 25kB
              Buffers: shared hit=5932 dirtied=1
              Worker 0:  Sort Method: quicksort  Memory: 25kB
              Worker 1:  Sort Method: quicksort  Memory: 25kB
              ->  Partial HashAggregate  (cost=4939.56..4939.77 rows=21 width=12) (actual time=5.512..5.513 rows
    =17 loops=3)
                    Group Key: h.property_type
                    Batches: 1  Memory Usage: 24kB
                    Buffers: shared hit=5914 dirtied=1
                    Worker 0:  Batches: 1  Memory Usage: 24kB
                    Worker 1:  Batches: 1  Memory Usage: 24kB
                    ->  Merge Join  (cost=2096.64..4789.56 rows=30000 width=4) (actual time=1.674..4.098 rows=24
    001 loops=3)
                          Merge Cond: (b.hotel_id = h.hotel_id)
                          Buffers: shared hit=5914 dirtied=1
                          ->  Parallel Index Only Scan using ix_bookings_hotel_dates on bookings b  (cost=0.29..
    2300.29 rows=30000 width=4) (actual time=0.020..1.027 rows=24001 loops=3)
                                Heap Fetches: 81
                                Buffers: shared hit=357 dirtied=1
                          ->  Sort  (cost=2096.35..2105.31 rows=3583 width=8) (actual time=1.631..1.719 rows=357
    5 loops=3)
                                Sort Key: h.hotel_id
                                Sort Method: quicksort  Memory: 208kB
                                Buffers: shared hit=5557
                                Worker 0:  Sort Method: quicksort  Memory: 208kB
                                Worker 1:  Sort Method: quicksort  Memory: 208kB
                                ->  Seq Scan on hotels h  (cost=0.00..1884.83 rows=3583 width=8) (actual time=0.
    011..1.269 rows=3583 loops=3)
                                      Buffers: shared hit=5547
```

```
Planning:
  Buffers: shared hit=13
Planning Time: 0.052 ms
Execution Time: 8.549 ms
```

operators: Aggregate, Gather Merge, Index Only Scan, Merge Join (Inner), Seq Scan, Sort, Sort Method: quicksort · execution 7.742 ms

Same JOIN, hash & merge disabled → (index) nested-loop join

Algorithms for JOIN › Nested-loop join

For each outer row, look up the matching inner rows (here through the hotels primary-key index).
setup: SET LOCAL enable_hashjoin = off; SET LOCAL enable_mergejoin = off

```
SELECT h.property_type, count(*) FROM bookings b JOIN hotels h ON h.hotel_id = b.hotel_id GROUP BY 1
```

```
HashAggregate  (cost=5178.06..5178.27 rows=21 width=12) (actual time=19.605..19.606 rows=21 loops=1)
  Group Key: h.property_type
  Batches: 1  Memory Usage: 24kB
  Buffers: shared hit=11639
  ->  Nested Loop  (cost=0.29..4818.06 rows=72000 width=4) (actual time=0.005..15.574 rows=72002 loops=1)
        Buffers: shared hit=11639
        ->  Seq Scan on bookings b  (cost=0.00..1616.00 rows=72000 width=4) (actual time=0.001..2.203 rows=72002
     loops=1)
              Buffers: shared hit=896
        ->  Memoize  (cost=0.29..0.41 rows=1 width=8) (actual time=0.000..0.000 rows=1 loops=72002)
              Cache Key: b.hotel_id
              Cache Mode: logical
              Hits: 68421  Misses: 3581  Evictions: 0  Overflows: 0  Memory Usage: 378kB
              Buffers: shared hit=10743
              ->  Index Scan using pk_hotels on hotels h  (cost=0.28..0.40 rows=1 width=8) (actual time=0.001..0
     .001 rows=1 loops=3581)
                    Index Cond: (hotel_id = b.hotel_id)
                    Buffers: shared hit=10743
Planning Time: 0.038 ms
Execution Time: 19.616 ms
```

operators: Aggregate, Index Scan, Memoize, Nested Loop (Inner), Seq Scan · execution 19.286 ms

ORDER BY bigger than memory → external merge sort on disk

Algorithms for external sorting

With only 64 kB of sort memory, PostgreSQL writes sorted runs to temporary files and merges them (look for "Sort Method: external merge Disk").
setup: SET LOCAL work_mem = '64kB'

```
SELECT booking_id, booked_at FROM bookings ORDER BY booked_at DESC
```

```
Sort  (cost=11120.86..11300.86 rows=72000 width=12) (actual time=20.131..22.295 rows=72002 loops=1)
  Sort Key: booked_at DESC
  Sort Method: external merge  Disk: 1896kB
  Buffers: shared hit=896, temp read=697 written=766
  ->  Seq Scan on bookings  (cost=0.00..1616.00 rows=72000 width=12) (actual time=0.001..4.945 rows=72002 loops=
    1)
        Buffers: shared hit=896
Planning:
  Buffers: shared hit=6
Planning Time: 0.018 ms
Execution Time: 24.342 ms
```

operators: Seq Scan, Sort, Sort Method: external merge · execution 24.057 ms

The same ORDER BY with enough memory → quicksort in RAM

Algorithms for external sorting (comparison)

Now the whole input fits: "Sort Method: quicksort Memory". Compare the time with plan.external_sort.
setup: SET LOCAL work_mem = '32MB'

```
SELECT booking_id, booked_at FROM bookings ORDER BY booked_at DESC
```

```
Sort  (cost=7424.86..7604.86 rows=72000 width=12) (actual time=9.819..11.313 rows=72002 loops=1)
  Sort Key: booked_at DESC
  Sort Method: quicksort  Memory: 5885kB
  Buffers: shared hit=896
```

```
  ->  Seq Scan on bookings  (cost=0.00..1616.00 rows=72000 width=12) (actual time=0.001..4.702 rows=72002 loops=
    1)
        Buffers: shared hit=896
Planning Time: 0.010 ms
Execution Time: 12.706 ms
```

operators: Seq Scan, Sort, Sort Method: quicksort · execution 12.559 ms

ORDER BY … LIMIT → top-N heapsort

Algorithms for sorting › Top-N

Only the best 10 rows are kept in a small heap; the input is never fully sorted.

```
SELECT booking_id, booked_at FROM bookings ORDER BY booked_at DESC LIMIT 10
```

```
Limit  (cost=3171.89..3171.92 rows=10 width=12) (actual time=8.985..8.986 rows=10 loops=1)
  Buffers: shared hit=896
  ->  Sort  (cost=3171.89..3351.89 rows=72000 width=12) (actual time=8.985..8.985 rows=10 loops=1)
        Sort Key: booked_at DESC
        Sort Method: top-N heapsort  Memory: 25kB
        Buffers: shared hit=896
        ->  Seq Scan on bookings  (cost=0.00..1616.00 rows=72000 width=12) (actual time=0.001..4.729 rows=72002
    loops=1)
              Buffers: shared hit=896
Planning Time: 0.012 ms
Execution Time: 8.989 ms
```

operators: Limit, Seq Scan, Sort, Sort Method: top-N heapsort · execution 8.963 ms

PROJECT with DISTINCT → hash aggregate (duplicate elimination)

Algorithms for PROJECT and set operations

Duplicate elimination: either read guest_id in sorted (index) order and drop repeats (Unique), or use a hash table (HashAggregate). The optimizer picks the cheaper one.

```
SELECT DISTINCT guest_id FROM bookings
```

```
HashAggregate  (cost=1796.00..1960.78 rows=16478 width=4) (actual time=7.591..8.335 rows=20456 loops=1)
  Group Key: guest_id
  Batches: 1  Memory Usage: 1809kB
  Buffers: shared hit=896
  ->  Seq Scan on bookings  (cost=0.00..1616.00 rows=72000 width=4) (actual time=0.001..2.031 rows=72002 loops=1
    )
        Buffers: shared hit=896
Planning Time: 0.015 ms
Execution Time: 8.748 ms
```

operators: Aggregate, Seq Scan · execution 8.805 ms

INTERSECT → hashed set operation

Algorithms for set operations

Both inputs are appended and a HashSetOp counts which rows came from both sides.

```
SELECT guest_id FROM bookings WHERE hotel_id = 1
INTERSECT
SELECT guest_id FROM bookings WHERE hotel_id = 3
```

```
HashSetOp Intersect  (cost=10.04..1413.44 rows=225 width=8) (actual time=0.301..0.304 rows=14 loops=1)
  Buffers: shared hit=635
  ->  Append  (cost=10.04..1411.38 rows=826 width=8) (actual time=0.025..0.265 rows=821 loops=1)
        Buffers: shared hit=635
        ->  Subquery Scan on "*SELECT* 2"  (cost=10.04..533.53 rows=226 width=8) (actual time=0.025..0.078 rows=
    247 loops=1)
              Buffers: shared hit=221
              ->  Bitmap Heap Scan on bookings  (cost=10.04..531.27 rows=226 width=4) (actual time=0.025..0.068
    rows=247 loops=1)
                    Recheck Cond: (hotel_id = 3)
                    Heap Blocks: exact=218
                    Buffers: shared hit=221
                    ->  Bitmap Index Scan on ix_bookings_hotel_dates  (cost=0.00..9.99 rows=226 width=0) (actual
     time=0.013..0.013 rows=247 loops=1)
                          Index Cond: (hotel_id = 3)
                          Buffers: shared hit=3
        ->  Subquery Scan on "*SELECT* 1"  (cost=16.94..873.72 rows=600 width=8) (actual time=0.048..0.156 rows=
```

```
    574 loops=1)
              Buffers: shared hit=414
              ->  Bitmap Heap Scan on bookings bookings_1  (cost=16.94..867.72 rows=600 width=4) (actual time=0.
    047..0.131 rows=574 loops=1)
                    Recheck Cond: (hotel_id = 1)
                    Heap Blocks: exact=410
                    Buffers: shared hit=414
                    ->  Bitmap Index Scan on ix_bookings_hotel_dates  (cost=0.00..16.79 rows=600 width=0) (actua
    l time=0.026..0.026 rows=574 loops=1)
                          Index Cond: (hotel_id = 1)
                          Buffers: shared hit=4
Planning Time: 0.036 ms
Execution Time: 0.313 ms
```

operators: Append, Bitmap Heap Scan, Bitmap Index Scan, SetOp, Subquery Scan · execution 0.29 ms

Pipelining — rows stream through operators without temp tables

Combining operations using pipelining

Scan → join → filter → limit pass rows one at a time. LIMIT stops the whole pipeline early: note "actual rows" far below table size.

```
SELECT b.booking_ref, h.hotel_name
FROM bookings b JOIN hotels h ON h.hotel_id = b.hotel_id
WHERE h.star_rating >= 4
LIMIT 5
```

```
Limit  (cost=0.29..6.24 rows=5 width=28) (actual time=0.012..0.040 rows=5 loops=1)
  Buffers: shared hit=64
  ->  Nested Loop  (cost=0.29..4826.77 rows=4059 width=28) (actual time=0.012..0.040 rows=5 loops=1)
        Buffers: shared hit=64
        ->  Seq Scan on bookings b  (cost=0.00..1616.00 rows=72000 width=13) (actual time=0.001..0.002 rows=21 l
    oops=1)
              Buffers: shared hit=1
        ->  Memoize  (cost=0.29..0.42 rows=1 width=23) (actual time=0.002..0.002 rows=0 loops=21)
              Cache Key: b.hotel_id
              Cache Mode: logical
              Hits: 0  Misses: 21  Evictions: 0  Overflows: 0  Memory Usage: 2kB
              Buffers: shared hit=63
              ->  Index Scan using pk_hotels on hotels h  (cost=0.28..0.41 rows=1 width=23) (actual time=0.002..
    0.002 rows=0 loops=21)
                    Index Cond: (hotel_id = b.hotel_id)
                    Filter: (star_rating >= 4)
                    Rows Removed by Filter: 1
                    Buffers: shared hit=63
Planning:
  Buffers: shared hit=13
Planning Time: 0.066 ms
Execution Time: 0.047 ms
```

operators: Index Scan, Limit, Memoize, Nested Loop (Inner), Seq Scan · execution 0.025 ms

Heuristic optimization — selection pushed below the join

Using heuristics in query optimization

We wrote the filter on cities after a 3-way join, but the planner applies σ city_name = 'Udaipur' first (on the cities scan), so the joins handle a few rows instead of thousands.

```
SELECT count(*)
FROM bookings b
JOIN hotels h ON h.hotel_id = b.hotel_id
JOIN cities c ON c.city_id = h.city_id
WHERE c.city_name = 'Udaipur'
```

```
Aggregate  (cost=49.12..49.13 rows=1 width=8) (actual time=0.225..0.225 rows=1 loops=1)
  Buffers: shared hit=129
  ->  Nested Loop  (cost=4.90..48.79 rows=133 width=0) (actual time=0.012..0.193 rows=1359 loops=1)
        Buffers: shared hit=129
        ->  Nested Loop  (cost=4.61..39.49 rows=7 width=4) (actual time=0.009..0.030 rows=41 loops=1)
              Buffers: shared hit=43
              ->  Index Scan using uq_cities_name_state on cities c  (cost=0.28..8.29 rows=1 width=4) (actual ti
    me=0.002..0.002 rows=1 loops=1)
                    Index Cond: ((city_name)::text = 'Udaipur'::text)
                    Buffers: shared hit=3
              ->  Bitmap Heap Scan on hotels h  (cost=4.33..31.13 rows=7 width=8) (actual time=0.006..0.024 rows
```

```
    =41 loops=1)
                  Recheck Cond: (city_id = c.city_id)
                  Heap Blocks: exact=38
                  Buffers: shared hit=40
                  ->  Bitmap Index Scan on ix_hotels_city  (cost=0.00..4.33 rows=7 width=0) (actual time=0.003
    ..0.003 rows=41 loops=1)
                        Index Cond: (city_id = c.city_id)
                        Buffers: shared hit=2
        ->  Index Only Scan using ix_bookings_hotel_dates on bookings b  (cost=0.29..1.12 rows=21 width=4) (actu
    al time=0.002..0.003 rows=33 loops=41)
              Index Cond: (hotel_id = h.hotel_id)
              Heap Fetches: 2
              Buffers: shared hit=86
Planning:
  Buffers: shared hit=25
Planning Time: 0.100 ms
Execution Time: 0.234 ms
```

operators: Aggregate, Bitmap Heap Scan, Bitmap Index Scan, Index Only Scan, Index Scan, Nested Loop (Inner) · execution 0.189 ms

Distributed fragments — only the relevant "site" is scanned

Unit 5 › Distributed DB (fragment pruning)

The query asks for SOUTH only, so the planner scans lab.site_south and skips the other five fragments.

```
SELECT count(*) FROM lab.bookings_by_region WHERE region = 'SOUTH'
```

```
Aggregate  (cost=564.57..564.58 rows=1 width=8) (actual time=2.274..2.274 rows=1 loops=1)
  Buffers: shared hit=225
  ->  Seq Scan on site_south bookings_by_region  (cost=0.00..507.98 rows=22638 width=0) (actual time=0.004..1.77
    8 rows=22638 loops=1)
        Filter: (region = 'SOUTH'::region_t)
        Buffers: shared hit=225
Planning:
  Buffers: shared hit=87
Planning Time: 0.994 ms
Execution Time: 2.277 ms
```

operators: Aggregate, Seq Scan · execution 1.532 ms

Full-text search → GIN (inverted) index

Unit 5 › Digital libraries (inverted index)

The GIN index maps each word stem to the hotels containing it, so the search does not read every description.

```
SELECT hotel_name FROM hotels WHERE search_doc @@ websearch_to_tsquery('english', 'heritage palace lake')
```

```
Bitmap Heap Scan on hotels  (cost=30.20..34.22 rows=1 width=19) (actual time=0.025..0.025 rows=2 loops=1)
  Recheck Cond: (search_doc @@ '''heritag'' & ''palac'' & ''lake'''::tsquery)
  Heap Blocks: exact=2
  Buffers: shared hit=9
  ->  Bitmap Index Scan on ix_hotels_search_doc  (cost=0.00..30.20 rows=1 width=0) (actual time=0.023..0.023 row
    s=2 loops=1)
        Index Cond: (search_doc @@ '''heritag'' & ''palac'' & ''lake'''::tsquery)
        Buffers: shared hit=7
Planning:
  Buffers: shared hit=1
Planning Time: 0.051 ms
Execution Time: 0.028 ms
```

operators: Bitmap Heap Scan, Bitmap Index Scan · execution 0.01 ms

11. Transactions, concurrency control and recovery

ACID: **Atomicity** — fn_book_stay inserts booking, rooms and payment together or not at all. **Consistency** — every constraint and trigger holds at commit (the capacity trigger is deferred to commit time). **Isolation** — row locks and MVCC snapshots keep concurrent bookings apart. **Durability** — the write-ahead log (WAL) is flushed before COMMIT returns. Each demonstration below was run with two real sessions while building this report.

Double booking race

t (ms)	session	step	result
10	A	BEGIN · fn_book_stay(guest 101, Maharani Suite) → locks the room FOR UPDATE	NULL
22	A	booking inserted · transaction still open, lock held	{'booking_ref': 'YRUVYQD3'}
227	B	BEGIN · fn_book_stay(guest 202, same room, same dates) → blocked, waiting for A's lock…	NULL
1533	A	COMMIT · lock released	NULL
1537	B	lock granted, availability re-checked → REJECTED	Only 0 room(s) of this type are free from 2027-02-15 to 2027-02-17
1562	—	clean-up: lab.purge_demo_bookings() removed the demo booking	1

SELECT … FOR UPDATE (an exclusive row lock, two-phase locking) makes B wait until A commits. B then re-checks availability and is refused: the last room can never be sold twice.

Deadlock

t (ms)	session	step	result
6	A	locked row 1	NULL
6	B	locked row 2	NULL
6	B	now asks for row 1 (held by the other session)…	NULL
6	A	now asks for row 2 (held by the other session)…	NULL
1008	B	ERROR: deadlock detected → chosen as victim, transaction aborted	Process 73220 waits for ShareLock on transaction 1853; blocked by process 73219. Process 73219 waits for ShareLock on transaction 1854; b…
1008	A	got row 2 once the victim was aborted	NULL

A waits for B and B waits for A: a cycle in the wait-for graph. After deadlock_timeout (1 s) PostgreSQL detects it and aborts one victim. fn_book_stay prevents this by always locking rooms in room_id order.

Isolation levels

READ COMMITTED

t (ms)	session	step	result
16	Reader	[READ COMMITTED] first read of qty	10
18	Writer	UPDATE qty = qty - 3 · COMMIT	NULL
18	Reader	second read inside the SAME transaction	7

REPEATABLE READ

t (ms)	session	step	result
9	Reader	[REPEATABLE READ] first read of qty	10
10	Writer	UPDATE qty = qty - 3 · COMMIT	NULL
10	Reader	second read inside the SAME transaction	10

READ COMMITTED sees the other transaction's committed change on the second read (non-repeatable read). REPEATABLE READ reads from a snapshot taken at its first query, so both reads agree.

Mvcc versions

t (ms)	session	step	result
7	A	row version before (xmin = id of the creating transaction, ctid = physical slot)	{'xmin': '1720', 'xmax': '1853', 'ctid': '(0,2)', 'qty': 4}
7	A	BEGIN · my transaction id	{'txid': 1859}
8	A	after UPDATE (uncommitted): a NEW version with my xmin at a new ctid	{'xmin': '1859', 'xmax': '0', 'ctid': '(0,34)', 'qty': 5}
8	B	meanwhile B still reads the OLD version — no dirty read, and B did not wait	{'xmin': '1720', 'xmax': '1859', 'ctid': '(0,2)', 'qty': 4}
8	A	ROLLBACK · the new version is never made visible	NULL

Multiversion concurrency control: a writer creates a new tuple version stamped with its transaction id (a timestamp), readers see the version valid for their snapshot. Readers never block writers, and uncommitted data is never visible.

Atomic rollback

t (ms)	session	step	result
7	A	bookings before	{'n': 72002}
18	A	step 1 ✓ booking header inserted (Charminar Courtyard)	NULL
19	A	step 2 ✗ room 1 is in Samudra Mahal, not Charminar Courtyard	Room 101 belongs to hotel 1, but booking YRVJ8D55 is at hotel 2
21	A	ROLLBACK · bookings after — step 1 was undone too	{'n': 72002}

Atomicity: a transaction is all-or-nothing. Step 1 succeeded, step 2 broke a rule, so the whole unit was undone and the database is exactly as it was.

Crash recovery

./yup recovery commits one row, leaves a second row in an open transaction, stops the server with 'pg_ctl stop -m immediate' (no checkpoint — like a power cut) and restarts it. The server log shows 'database system was not properly shut down; automatic recovery in progress' and 'redo starts at …'. Afterwards the committed row is present (durability) and the uncommitted row is absent (atomicity): REDO replays the WAL, and a transaction without a commit record is never visible.

12. Distributed, object-oriented, multimedia databases, data mining and digital libraries

Horizontal fragmentation — bookings stored per region "site"

[dist.horizontal]

RA / RC: bookings = site_north ∪ site_south ∪ site_east ∪ site_west ∪ site_central ∪ site_northeast

Each fragment holds the rows of one region. The union of all fragments is the whole relation (completeness), and no row is in two fragments (disjointness).

```
SELECT tableoid::regclass AS fragment, count(*) AS rows,
       min(check_in) AS first_stay, max(check_in) AS last_stay
FROM lab.bookings_by_region
GROUP BY 1 ORDER BY rows DESC
```

fragment	rows	first_stay	last_stay
lab.site_west	22745	2025-01-01	2027-03-24
lab.site_south	22638	2025-01-01	2027-03-24
lab.site_north	19281	2025-01-01	2027-03-24
lab.site_east	4280	2025-01-01	2027-03-24
lab.site_central	2146	2025-01-01	2027-03-23
lab.site_northeast	910	2025-01-01	2027-03-19

Vertical fragmentation — guest columns at two sites, rejoined on the key [dist.vertical]

RA / RC: guests = guest_contact ⋈ guest_profile

Each fragment keeps the primary key, so the natural join reconstructs the original rows exactly (lossless).

```
SELECT (SELECT count(*) FROM guests) AS original_rows,
       (SELECT count(*) FROM lab.guest_contact_frag c JOIN lab.guest_profile_frag p USING (guest_id)) AS rejoine
    d_rows
```

original_rows	rejoined_rows
26002	26000

Inheritance — querying the parent class returns all subclasses

[oodb.inheritance]

RA / RC: —

lab.heritage_property and lab.floating_property INHERIT lab.property. tableoid::regclass tells which "class" each object belongs to.

```
SELECT tableoid::regclass::text AS stored_in, count(*) AS objects
FROM lab.property GROUP BY 1
UNION ALL
SELECT 'ONLY lab.property', count(*) FROM ONLY lab.property
```

stored_in	objects
lab.property	2
lab.heritage_property	14
lab.floating_property	80
ONLY lab.property	2

Composite type and a method on it [oodb.composite_type]

RA / RC: —

address is an object with its own attributes (address).city; lab.label(p) behaves like a method.

```
SELECT p.name, (p.address).city AS city, (p.address).state AS state, p.era, lab.label(p) AS label
```

```
FROM lab.heritage_property p
ORDER BY (p.address).state, p.name
LIMIT 12
```

name	city	state	era	label
Balaram Palace Resort	Chitrasani	Gujarat	Royal	Balaram Palace Resort — Chitrasani, Gujarat
Spice Warehouse Fort Kochi	Kochi	Kerala	Colonial / Mughal	Spice Warehouse Fort Kochi — Kochi, Kerala
Samudra Mahal	Mumbai	Maharashtra	Colonial / Mughal	Samudra Mahal — Mumbai, Maharashtra
Amber Haveli	Jaipur	Rajasthan	Colonial / Mughal	Amber Haveli — Jaipur, Rajasthan
Fort Dhariyawad	Pratapgarh	Rajasthan	Royal	Fort Dhariyawad — Pratapgarh, Rajasthan
Fort Pokaran	Jaisalmer	Rajasthan	Royal	Fort Pokaran — Jaisalmer, Rajasthan
Hotel Raj Niwas	Udaipur	Rajasthan	Royal	Hotel Raj Niwas — Udaipur, Rajasthan
Hotel Sikar Haveli-Heritage Home	Jaipur	Rajasthan	Royal	Hotel Sikar Haveli-Heritage Home — Jaipur, Rajasthan

… 4 more row(s) not shown.

Multimedia — SVG posters stored as binary (bytea) in the database

[media.posters]

RA / RC: —

The content column holds the image bytes; metadata (title, MIME type, size) sits alongside. The API serves them at /api/media/{id}.

```
SELECT media_id, hotel_id, title, mime_type, bytes
FROM lab.hotel_media ORDER BY media_id
```

media_id	hotel_id	title	mime_type	bytes
1	15	Solang Snowline Camp — poster	image/svg+xml	551
2	6	Vembanad Kettuvallam — poster	image/svg+xml	553
3	10	Sonar Qila Haveli — poster	image/svg+xml	551
4	9	Dal Shikara House — poster	image/svg+xml	550
5	2	Charminar Courtyard — poster	image/svg+xml	553
6	1	Samudra Mahal — poster	image/svg+xml	544
7	18	Rue Romain Villa — poster	image/svg+xml	552
8	17	Ganga Kinara Retreat — poster	image/svg+xml	554

… 10 more row(s) not shown.

Association rules — "hotels with a Spa usually also have …"

[mining.amenity_rules]

RA / RC: — (market-basket analysis)

support = P(A ∧ B); confidence = P(B | A). Computed by a self-join of hotel_amenities on hotel_id.

```
WITH n AS (SELECT count(DISTINCT hotel_id)::numeric AS hotels FROM hotel_amenities),
```

```
pairs AS (
    SELECT a.amenity_id AS a_id, b.amenity_id AS b_id, count(*) AS both_count
    FROM hotel_amenities a JOIN hotel_amenities b ON a.hotel_id = b.hotel_id AND a.amenity_id <> b.amenity_id
    GROUP BY 1, 2
), single AS (
    SELECT amenity_id, count(*) AS cnt FROM hotel_amenities GROUP BY amenity_id
)
SELECT am_a.amenity_name AS if_hotel_has, am_b.amenity_name AS then_also_has,
       round(p.both_count / n.hotels, 3) AS support,
       round(p.both_count::numeric / s.cnt, 3) AS confidence
FROM pairs p
JOIN single s ON s.amenity_id = p.a_id
JOIN amenities am_a ON am_a.amenity_id = p.a_id
JOIN amenities am_b ON am_b.amenity_id = p.b_id
CROSS JOIN n
WHERE am_a.amenity_name = 'Spa' AND s.cnt >= 20
ORDER BY confidence DESC
LIMIT 10
```

if_hotel_has	then_also_has	support	confidence
Spa	Room Service 24x7	0.13	0.94
Spa	Parking	0.13	0.92
Spa	Laundry	0.12	0.89
Spa	Front Desk 24x7	0.12	0.88
Spa	Restaurant	0.12	0.83
Spa	Car Rental	0.11	0.82
Spa	Doctor on Call	0.11	0.79
Spa	Power Backup	0.10	0.72

… 2 more row(s) not shown.

Travel patterns — which two cities are most often visited by the same guest [mining.city_pairs]

RA / RC: ρ x(trips) ⋈ x.guest = y.guest ∧ x.city < y.city ρ y(trips)

A self-join on guest_id finds every pair of cities one guest has stayed in.

```
WITH trips AS (
    SELECT DISTINCT b.guest_id, c.city_name
    FROM bookings b JOIN hotels h USING (hotel_id) JOIN cities c USING (city_id)
    WHERE b.status = 'COMPLETED'
)
SELECT x.city_name AS city_a, y.city_name AS city_b, count(*) AS shared_guests
FROM trips x JOIN trips y ON x.guest_id = y.guest_id AND x.city_name < y.city_name
GROUP BY 1, 2
ORDER BY shared_guests DESC
LIMIT 10
```

city_a	city_b	shared_guests
Bangalore	Delhi	1039
Bangalore	Goa	776
Delhi	Goa	773
Bangalore	Chennai	628
Chennai	Delhi	621
Delhi	Mumbai	602
Bangalore	Mumbai	587
Bangalore	Hyderabad	552

… 2 more row(s) not shown.

Digital libraries / information retrieval: hotels.search_doc is a generated tsvector (weighted name, locality, tagline and description) with a GIN inverted index; 'quiet hill stay' is matched by word stems and ranked with ts_rank (section 9). A trigram index gives typo-tolerant name

search.

13. Interfaces: API, CLI and website

method	endpoint	SQL / object used
GET	/api/destinations?q=	UNION of cities, landmarks, hotels with trigram similarity
GET	/api/search?…	joins + full-text + division (must-have amenities) + fn_available_rooms
GET	/api/hotels/{id}	hotels ⋈ cities ⋈ states, amenities, landmarks, room types, reviews
GET	/api/hotels/{id}/availability	fn_available_rooms, fn_stay_total
GET	/api/quote	fn_quote_stay (night-by-night price)
POST	/api/auth/otp, /api/auth/verify	lookup by candidate key guests.phone (OTP mocked)
POST	/api/guests	INSERT guests (domains, UNIQUE, CHECK age)
POST	/api/bookings	fn_book_stay (locks, triggers, payment) in one transaction
POST	/api/bookings/{ref}/cancel	CALL sp_cancel_booking
POST	/api/bookings/{ref}/review	INSERT reviews (trg_reviews_guard, trg_reviews_rollup)
GET/POST	/api/lab/…	catalog browser, named queries, EXPLAIN, why-not, concurrency

13.1 CLI

```
./yup tables                        15 tables, rows, PK, FK/UNIQUE/CHECK counts
./yup show hotels -w "star_rating = 5" -n 10
./yup describe bookings             columns, constraints, referenced-by, indexes, triggers
./yup keys                          primary, candidate and foreign keys
./yup queries -t join               list named queries        ./yup q joins.full_outer
./yup plans                         ./yup explain plan.external_sort
./yup why                           run all 16 constraint scenarios   ./yup why why.double_booking
./yup txn deadlock                  (double_booking_race | isolation_levels | mvcc_versions | atomic_rollback)
./yup quote 1 2026-12-24 2026-12-26
./yup find "gateway"                ./yup sql "SELECT count(*) FROM bookings"
./yup recovery                      crash-recovery demonstration
```

13.2 Website

Home (search by city, landmark or mood; signature stays; festival price calendar) → Search results (filters in the URL, compare up to three stays) → Hotel page (story, neighbourhood with walking times, rooms priced for your dates, night-by-night breakdown, reviews) → Checkout (phone + OTP sign-in, UPI first, honest price) → Confirmation (booking reference) → My trips (cancel with refund preview, write a review) → Under the hood (tables, query library, plans, why-not, transactions). English / □□□□□ and light / dark themes; motion respects the reduce-motion setting; all imagery is lightweight SVG drawn in code.

14. How to run

```
scripts/db.sh start                         # PostgreSQL on port 5433
scripts/db.sh build                         # (re)create the database from 01-dataset/processed (~3 min)
cd 03-backend && ~/.yupreserved/venv/bin/uvicorn app.main:app --port 8000
cd 05-frontend && npm run dev               # http://localhost:5173
./yup tables                                # CLI
python3 01-dataset/pipeline/build_dataset.py            # regenerate the dataset (optional)
~/.yupreserved/venv/bin/python 06-docs/report/build_report.py   # rebuild this PDF
```

Part C · Viva questions and answers

Design and ER model

Why did you choose a hotel reservation system?

It needs almost every DBMS idea for real reasons: many-to-many relationships (hotels ↔ amenities, bookings ↔ rooms), a weak entity (rooms), date-range rules no CHECK can express (double booking), money that must be exact, and concurrency (two people booking the last room).

Why exactly these 15 tables?

Each is one entity or one M:N relationship of the ER model: 12 entities (states, cities, landmarks, hotels, amenities, room_types, rooms, seasons, guests, bookings, payments, reviews) and 3 M:N relationship tables (hotel_amenities, hotel_landmarks, booking_rooms). Nothing is stored twice except controlled derived values.

Why is ROOM a weak entity?

A room number like '101' means nothing on its own — every hotel has a room 101. It is identified only together with its hotel: room_number is a **partial key**, and (hotel_id, room_number) identifies the room. In the table we keep that as a UNIQUE candidate key and add a surrogate room_id as primary key.

What is the cardinality of bookings and rooms?

M:N. A booking can hold several rooms (a family books 2), and a room appears in many bookings over time. Hence booking_rooms with composite PK (booking_id, room_id) and the relationship attributes room_subtotal and room_tax.

Where is a 1:1 relationship?

bookings ↔ reviews: reviews.booking_id is a foreign key AND UNIQUE, so a stay has at most one review.

Where is a recursive relationship?

guests.referred_by references guests.guest_id ('refers'). The self join and the recursive CTE (sub.recursive) use it.

Which attributes are derived and how are they handled?

bookings.nights is a GENERATED column (check_out − check_in); age is computed in v_guest_directory; hotels.avg_rating and review_count are stored for speed and kept correct by trigger trg_reviews_rollup.

Keys

Why is state_code a natural primary key but hotels use a surrogate key?

State codes (MH, RJ) are short, stable and universally known, so they make good natural keys. Hotel names are not unique and can change; source_ref is long text. A surrogate integer is small, never changes and keeps joins fast.

Name a super key, a candidate key and an alternate key in guests.

Super key: {guest_id, phone}. Candidate keys: {guest_id}, {phone}, {email}, {id_type, id_number}. Primary key: guest_id. Alternate keys: phone, email and (id_type, id_number), each enforced with UNIQUE.

Why is (id_type, id_number) a composite key?

Neither column is unique alone — two guests can both use PAN, and a passport and a PAN could share digits — but the pair identifies one document.

Why does rooms have UNIQUE (room_type_id, hotel_id) in room_types?

So that rooms can declare the composite foreign key (room_type_id, hotel_id) → room_types. That FK guarantees a room's type belongs to the same hotel as the room; a single-column FK could not.

Why does booking_rooms NOT contain hotel_id?

With key (booking_id, room_id), hotel_id would depend on booking_id alone — a partial dependency (2NF violation). The trigger trg_booking_rooms_guard checks that the room's hotel equals the booking's hotel instead.

Why ON DELETE RESTRICT for bookings → hotels but CASCADE for room_types → hotels?

Bookings and payments are history and money: deleting a hotel must not silently erase them. Room types, rooms and amenity links are part of the hotel (composition), so they go with it — but only if no booking still uses a room.

Constraints and triggers

Why can't I enter a check-out date before the check-in date?

CHECK constraint ck_bookings_dates (check_out > check_in). The DBMS rejects the row whatever program sends it.

Why is double booking prevented by a trigger and not a CHECK?

A CHECK sees only the row being inserted. 'Is this room already taken for an overlapping range?' needs OTHER rows of booking_rooms and bookings, so it is a BEFORE INSERT trigger (trg_booking_rooms_guard).

How do you test whether two stays overlap?

Stays are half-open ranges [check_in, check_out). Two ranges [a,b) and [c,d) overlap exactly when a < d AND c < b. So checking out and another guest checking in on the same day is allowed.

What is a deferred constraint trigger and why use one?

trg_booking_capacity is DEFERRABLE INITIALLY DEFERRED: it runs at COMMIT. A booking for 5 adults inserts its rooms one by one; checking after the first room would wrongly fail. At commit all rooms exist and capacity can be judged.

What does the EXCLUDE constraint on seasons do?

It uses a GiST index on daterange(start_date, end_date) with the overlap operator so that no two seasons share a night; otherwise a night would have two multipliers.

Your photo CHECK first rejected cities without a photo. Why?

CHECK (photo_url ~ '^https://' AND photo_credit IS NOT NULL): for a NULL photo the regex gives UNKNOWN and the credit test gives FALSE; UNKNOWN AND FALSE = FALSE, so the row was rejected (a CHECK fails only on FALSE, passes on TRUE or UNKNOWN). The fix: CHECK (photo_url IS NULL OR (photo_url ~ '^https://' AND photo_credit IS NOT NULL)).

How is the OTP stored securely?

Never in plain text: auth.otp_challenges keeps an HMAC-SHA256 of (phone, code) with a server secret, an expiry time and an attempt counter (CHECK attempts BETWEEN 0 AND 5). Verification locks the row FOR UPDATE and compares hashes in constant time.

What is a domain? Give an example.

A named type with a rule: indian_mobile is varchar(10) CHECK (VALUE ~ '^[6-9][0-9]{9}$'). Every phone column uses it.

Why can't a completed booking go back to CONFIRMED?

trg_bookings_status implements a state machine: PENDING → CONFIRMED → CHECKED_IN → COMPLETED, plus CANCELLED / NO_SHOW. Any other change raises an error, so history cannot be rewritten.

Normalization

Where did 1NF fail in your raw data?

hotel_facilities held 'Parking|Room Service|Wi-Fi|...' in one cell — a repeating group. lab.nf1_listing_facility makes one row per (hotel, facility); the final design is hotel_amenities.

Show a partial dependency you removed (2NF).

In nf1_listing_facility(property_id, facility, property_name, city, state) with key (property_id, facility), property_id → property_name, city, state depends on part of the key. Split into nf2_property and nf2_property_facility.

Show a transitive dependency you removed (3NF).

property_id → city → state. The state was repeated for every hotel of a city; nf3_city(city, state) removes it. In the final schema: hotels.city_id → cities.state_code.

How do you prove the decomposition is lossless?

Query nf.third rejoins the 3NF tables and compares with the 1NF table using EXCEPT: zero rows lost, zero spurious. Theory: each split shares a key of one side (R1 ∩ R2 → R1 or R1 ∩ R2 → R2).

What happens if you decompose on an FD that does not hold?

The join is lossy. On the raw names, city → state is false (Hyderabad under two states), and the 3NF re-join gave 357 lost/spurious rows. City names are not unique either (Aurangabad: Maharashtra and Bihar), which is why cities has the surrogate key city_id and the candidate key (city_name, state_code).

Is every table in BCNF?

All except rooms, deliberately. room_type_id → hotel_id has a non-superkey determinant. rooms is still 3NF because hotel_id is prime. A BCNF split would lose the dependency (hotel_id, room_number) → room_id, so we keep 3NF and a composite FK.

What is a multivalued dependency here (4NF)?

hotel ↠ amenity and hotel ↠ landmark are independent. Storing both in one table forces every amenity to pair with every landmark (nf.fourth shows the blow-up). 4NF keeps them in hotel_amenities and hotel_landmarks.

What is a join dependency (5NF)?

lab.jd_offer(hotel, meal_plan, channel) equals the join of its three binary projections, but the join of any two creates spurious rows (nf.fifth). That 3-way join dependency is what 5NF is about.

Why do you store room_subtotal if prices can be calculated?

It is not derivable later: base prices and seasons can change, but the price agreed at booking time is a fact of that booking. It depends on the whole key (booking_id, room_id).

Queries, algebra and calculus

How do you retrieve the price for two nights from date X to date Y?

SELECT * FROM fn_quote_stay(room_type_id, X, Y): a CROSS JOIN of the room type with generate_series(nights), a THETA LEFT JOIN to seasons (night BETWEEN start AND end), × weekend factor, then the GST slab per night.

Which join returns states that have no hotels?

LEFT OUTER JOIN states → cities → hotels (joins.left_outer): Lakshadweep and Manipur appear with 0.

Write division in SQL.

Hotels that have ALL of {Pool, Spa, Wi-Fi}: double NOT EXISTS — 'no required amenity that the hotel lacks' (division.not_exists), or GROUP BY hotel HAVING count(DISTINCT amenity) = 3 (division.count).

What is the difference between TRC and DRC?

In tuple relational calculus variables range over whole tuples ({g | guests(g) ∧ ∃b …}); in domain relational calculus they range over attribute values ({⟨n, s⟩ | ∃i, t, c hotels(i, n, t, s, c) ∧ …}). SQL's EXISTS is TRC.

Semi-join vs inner join?

An inner join repeats a hotel once per matching review; a semi-join (EXISTS) returns each hotel once if any match exists.

Why NOT EXISTS and not NOT IN for the anti-join?

If the subquery returns a NULL, x NOT IN (…) is UNKNOWN for every row and returns nothing; NOT EXISTS is NULL-safe.

Query processing

Which join algorithm does PostgreSQL use for bookings ⋈ hotels and why?

Hash join: build a hash table on the smaller input (3,583 hotels) and probe it with 72,000 bookings — one pass over each, O(|R| + |S|). With hash joins disabled it chooses a sort-merge join, and with both disabled a nested loop.

Show external sorting.

plan.external_sort sets work_mem = 64 kB; the plan reports 'Sort Method: external merge Disk: … kB' — sorted runs written to temp files and merged. With 32 MB it becomes an in-memory quicksort.

What is pipelining?

Operators pass tuples to their parent one at a time without materialising intermediate tables. In plan.pipelining LIMIT 5 stops the scan and join after a handful of rows.

Give an example of heuristic optimization.

Selection pushdown: in plan.heuristic_pushdown the filter city_name = 'Udaipur' is applied on the cities scan before the joins, so the joins process a few rows instead of thousands.

Why a GIN index on search_doc?

A GIN index is an inverted index (word → hotels containing it), so full-text search does not read every description.

Transactions, concurrency and recovery

What happens if two guests book the last room at the same time?

fn_book_stay locks the room rows with SELECT … FOR UPDATE. The second transaction waits; when the first commits, the second re-checks availability and is refused ('Only 0 room(s) … are free'). The race demo shows the 1.3 s wait.

How do you avoid deadlocks in booking?

Rooms are always locked in room_id order, so two transactions can never hold locks in opposite orders. The deadlock demo shows what happens otherwise: PostgreSQL detects the cycle after deadlock_timeout and aborts one victim.

Is PostgreSQL lock-based or timestamp-based?

Both ideas: writers take row locks (two-phase locking), while reads use MVCC — each row version carries transaction ids (xmin / xmax), a form of timestamp, and each statement or transaction reads a snapshot. Readers never block writers.

Show a non-repeatable read.

In READ COMMITTED a second read in the same transaction sees another session's committed update; in REPEATABLE READ both reads return the same value (txn isolation_levels).

How does the database recover from a crash?

Write-ahead logging: every change is written to the WAL before it is applied, and COMMIT waits for the WAL flush. After a crash PostgreSQL REDOes the log from the last checkpoint; transactions without a commit record stay invisible. ./yup recovery demonstrates it.

Why is fn_book_stay SECURITY DEFINER?

The app role yup_app may not UPDATE or lock rooms directly. The function runs with its owner's rights, so the only way the app can lock rooms is through this one audited booking routine — least privilege.

Dataset, system and Unit 5

Where does your data come from and is it real?

Hotel listings are real: the Goibibo dataset (4,000 Indian properties, Kaggle/PromptCloud). Prices, guests, bookings and reviews are simulated with a fixed seed because no public dataset contains them; the 18 Signature hotels are original. All of this is documented in 01-dataset/README.md.

Why did you choose PostgreSQL over MySQL?

The installed MySQL 5.7 ignores CHECK constraints and lacks FULL OUTER JOIN, INTERSECT/EXCEPT, CTEs and window functions. PostgreSQL also gives EXCLUDE, deferred constraints, inheritance, partitioning and full-text search.

How is a distributed database shown?

lab.bookings_by_region is horizontally fragmented by region into six partitions ('sites'); the planner prunes fragments (plan.partition_pruning). Vertical fragments of guests rejoin losslessly on the key.

How is an object-oriented database shown?

A composite type lab.address_t (an object with attributes), tables that INHERIT from lab.property (IS-A), and a function lab.label(p) acting as a method.

How is a multimedia database shown?

lab.hotel_media stores SVG posters as bytea with MIME type and size; the API streams them and the hotel page displays them.

Where is data mining?

Association rules on amenities (support and confidence via a self-join), co-visited city pairs, ROLLUP / CUBE and window functions.

Part D · Syllabus coverage

Unit 1 · Introduction

topic	where it is in YupReserved
Concept and goals of DBMS	Part B §1: data integrity, sharing, security, independence achieved by PostgreSQL
Database languages	DDL (schema/*.sql), DML (seed, queries), DCL (07_roles.sql), TCL (BEGIN/COMMIT/ROLLBACK in fn_book_stay, lab)
Database users	roles postgres (DBA), yup_app (application), yup_analyst (end user via views), yup_lab
Database abstraction	physical (files, indexes), logical (15 tables), view level (v_hotel_cards, v_guest_directory)
Database architecture	three-tier architecture (Part B §2); client–server PostgreSQL
Relational model and constraints	02_tables.sql: domain, key, entity-integrity, referential-integrity, CHECK, EXCLUDE
ER model, relationship sets, keys, mapping, design	Part B §4: Chen diagram, weak entity, cardinalities, ER → relational mapping table

Unit 2 · Relational model

topic	where it is in YupReserved
The catalog	catalog.* queries (pg_catalog / information_schema), ./yup describe, Lab 'Tables' tab
Types	ENUM types and DOMAINs (01_types.sql), composite type lab.address_t
Keys	catalog.primary_keys / candidate_keys / foreign_keys; Part B §5
Relational algebra (fundamental + additional operations)	σ π ρ ∪ − × in basics.*, sets.*, joins.cross; ⋈ ÷ ∩ γ in joins.*, division.*, agg.*
Tuple and domain relational calculus	calculus.trc, calculus.drc, division.not_exists
SQL fundamentals	66 named queries: predicates, CASE, subqueries, CTEs, recursion, window functions
Integrity	constraints + 16 why-not scenarios
Triggers	8 triggers incl. a deferred constraint trigger (04_triggers.sql)
Views	5 views incl. updatable WITH CHECK OPTION and a materialized view

Unit 3 · Relational database design

topic	where it is in YupReserved
Normalization using functional dependencies	Part B §7: FD table; UNF → 1NF → 2NF → 3NF on the real data (lab.nf*)
BCNF and dependency preservation	nf.bcnf_rooms — rooms deliberately 3NF
Multivalued dependency (4NF)	nf.fourth (lab.mvd_hotel_amenity_landmark)
Join dependency (5NF)	nf.fifth (lab.jd_offer)
Lossless decomposition	nf.third and dist.vertical

Unit 4 · Query processing, optimization and tuning

topic	where it is in YupReserved
Translating SQL into relational algebra	every named query carries its RA form; EXPLAIN shows the operator tree
External sorting	plan.external_sort vs plan.in_memory_sort, plan.top_n

topic	where it is in YupReserved
SELECT algorithms	plan.seq_scan, plan.index_scan, plan.bitmap_scan
JOIN algorithms	plan.hash_join, plan.merge_join, plan.nested_loop
PROJECT and set operations	plan.project_distinct, plan.set_operation
Pipelining	plan.pipelining
Heuristic optimization	plan.heuristic_pushdown
Database tuning	05_indexes.sql: FK, composite, partial, GIN, trigram and expression indexes; laptop-tuned settings

Unit 5 · Transactions, concurrency, recovery and new applications

topic	where it is in YupReserved
Transaction processing concepts	ACID in fn_book_stay; txn atomic_rollback
Lock-based protocols	SELECT … FOR UPDATE; txn double_booking_race
Timestamp-based protocols	MVCC transaction ids xmin/xmax; txn mvcc_versions
Deadlock handling	txn deadlock (detection + victim); ordered locking prevents it in bookings
Crash recovery	WAL; ./yup recovery
Distributed database	horizontal + vertical fragmentation (lab.bookings_by_region, guest fragments)
Object-oriented database	composite types, table inheritance, method function (lab.*)
Multimedia database	lab.hotel_media (bytea), served at /api/lab/media/{id}
Data mining	mining.amenity_rules, mining.city_pairs, agg.rollup, agg.cube, sub.window
Digital libraries	full-text search (tsvector + GIN), app.mood_search, trigram search